

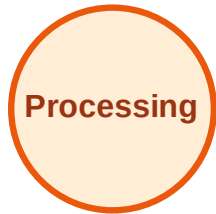
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



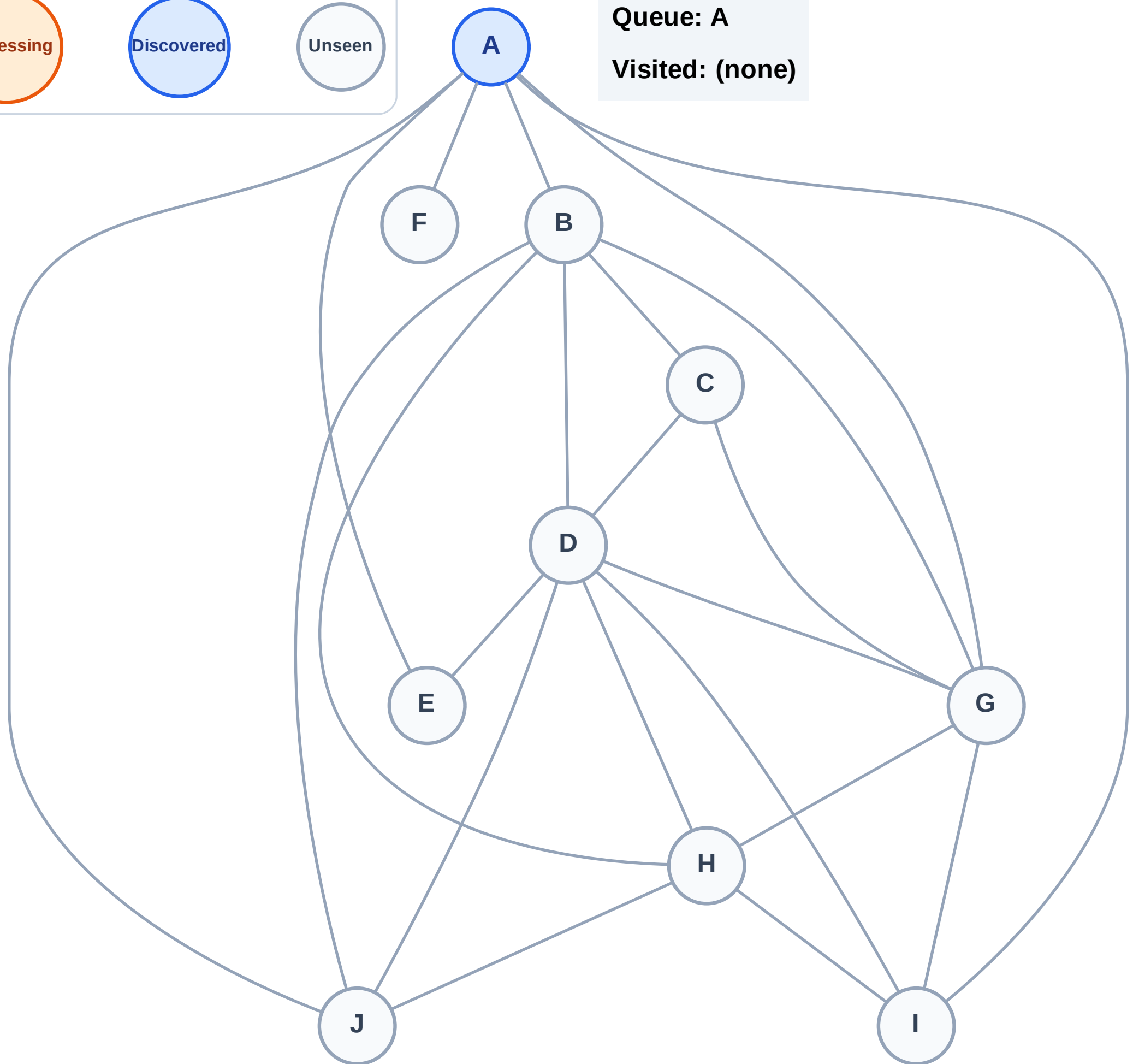
Discovered



Unseen

Queue: A

Visited: (none)



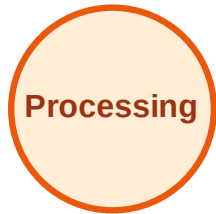
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



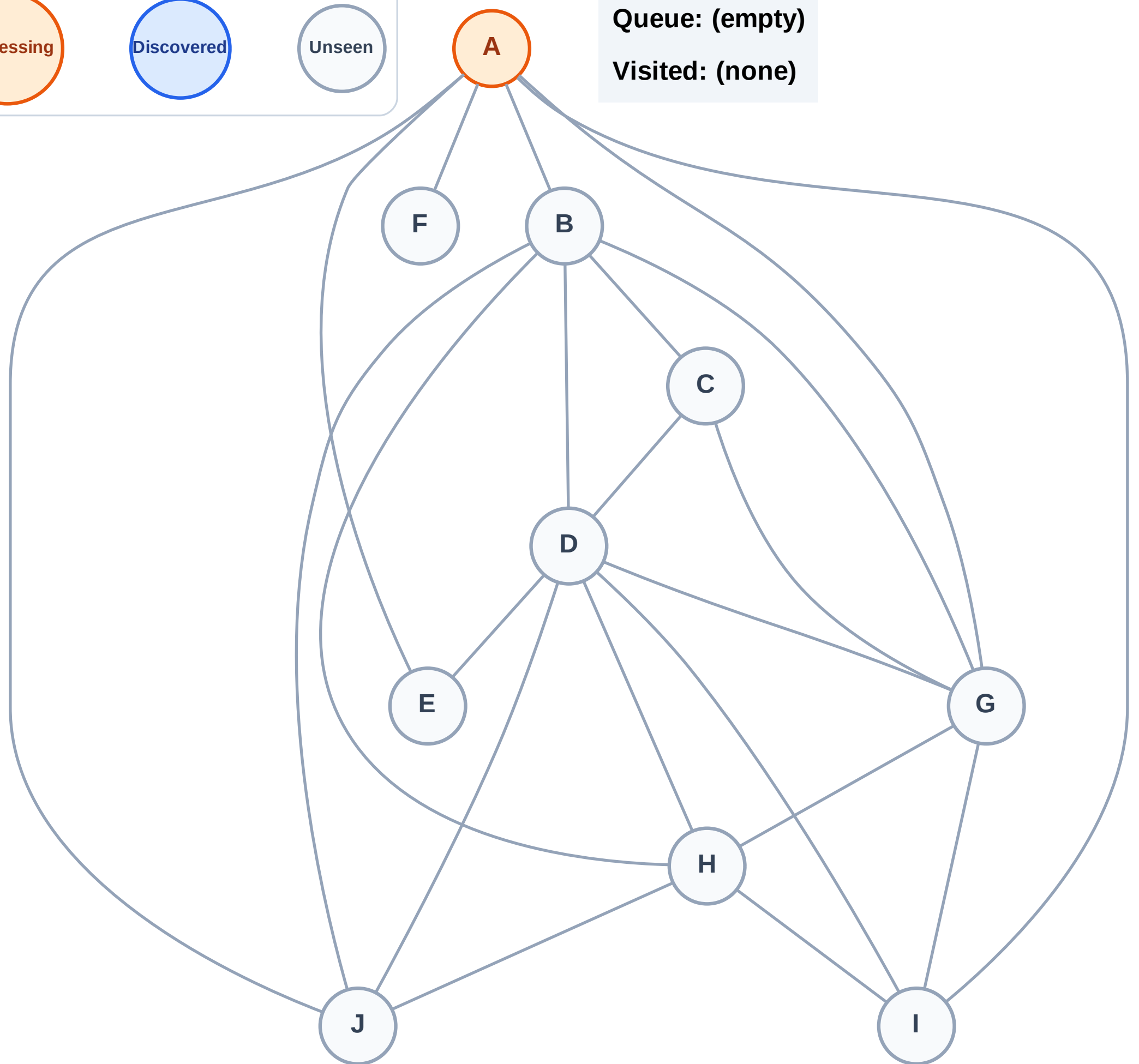
Discovered



Unseen

Queue: (empty)

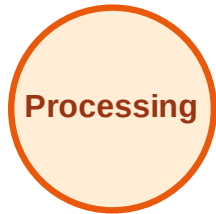
Visited: (none)



BFS Traversal

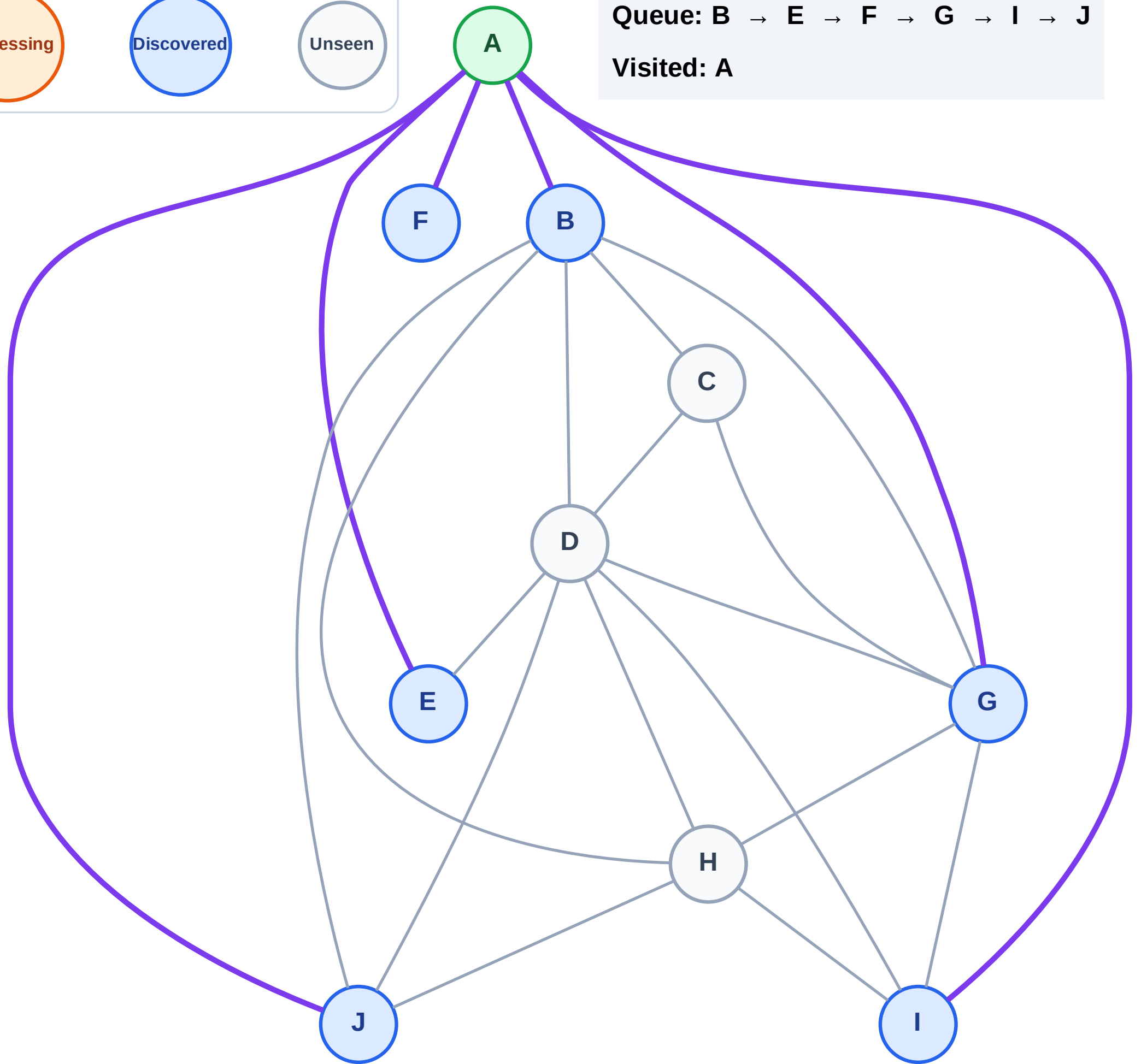
Step 3: Discover B, E, F, G, I, J from A

Legend



Queue: B → E → F → G → I → J

Visited: A



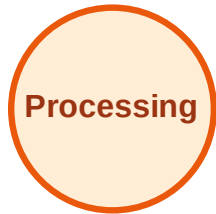
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



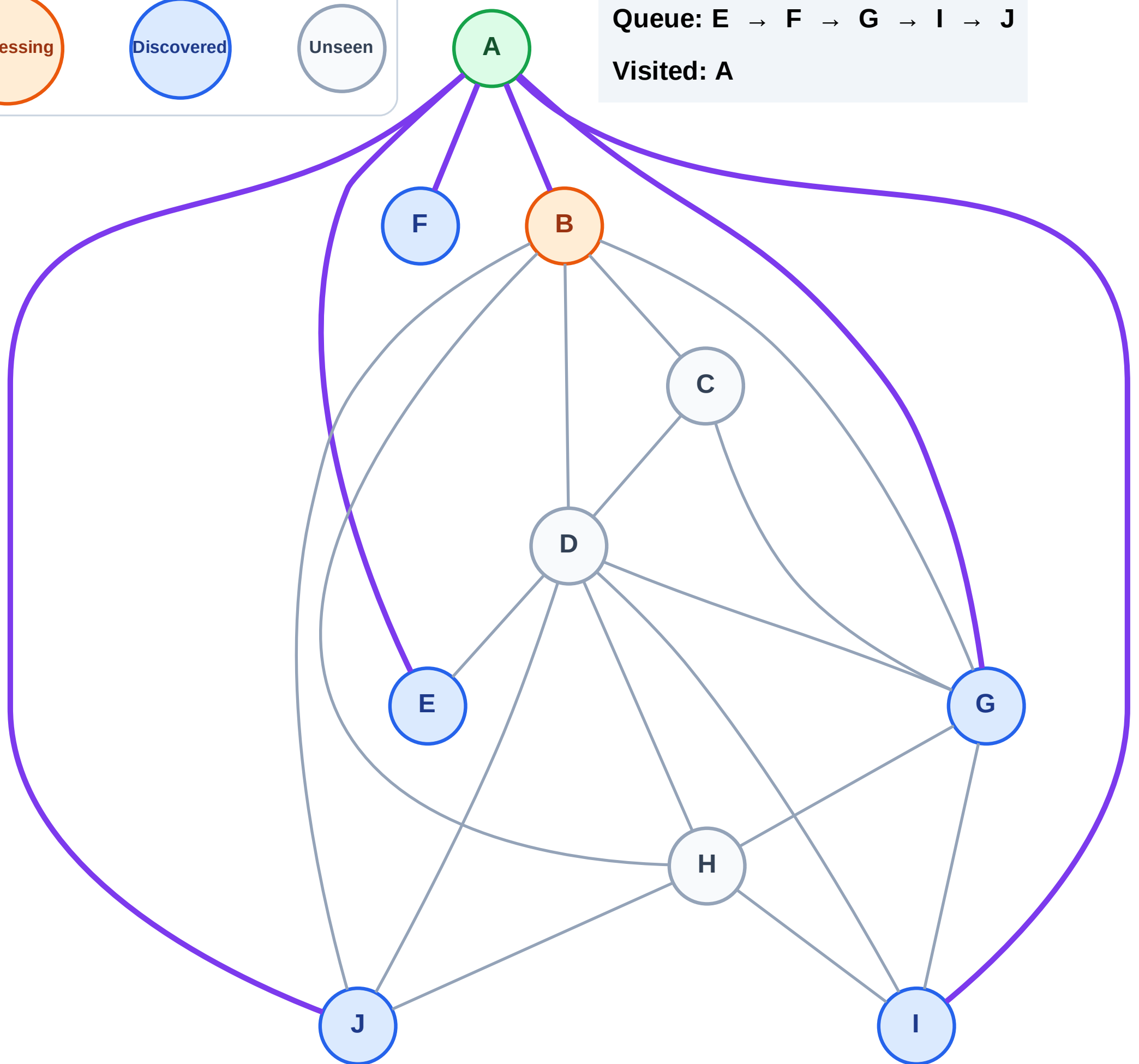
Discovered



Unseen

Queue: E → F → G → I → J

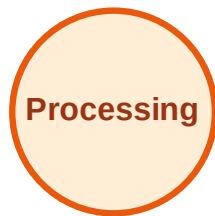
Visited: A



BFS Traversal

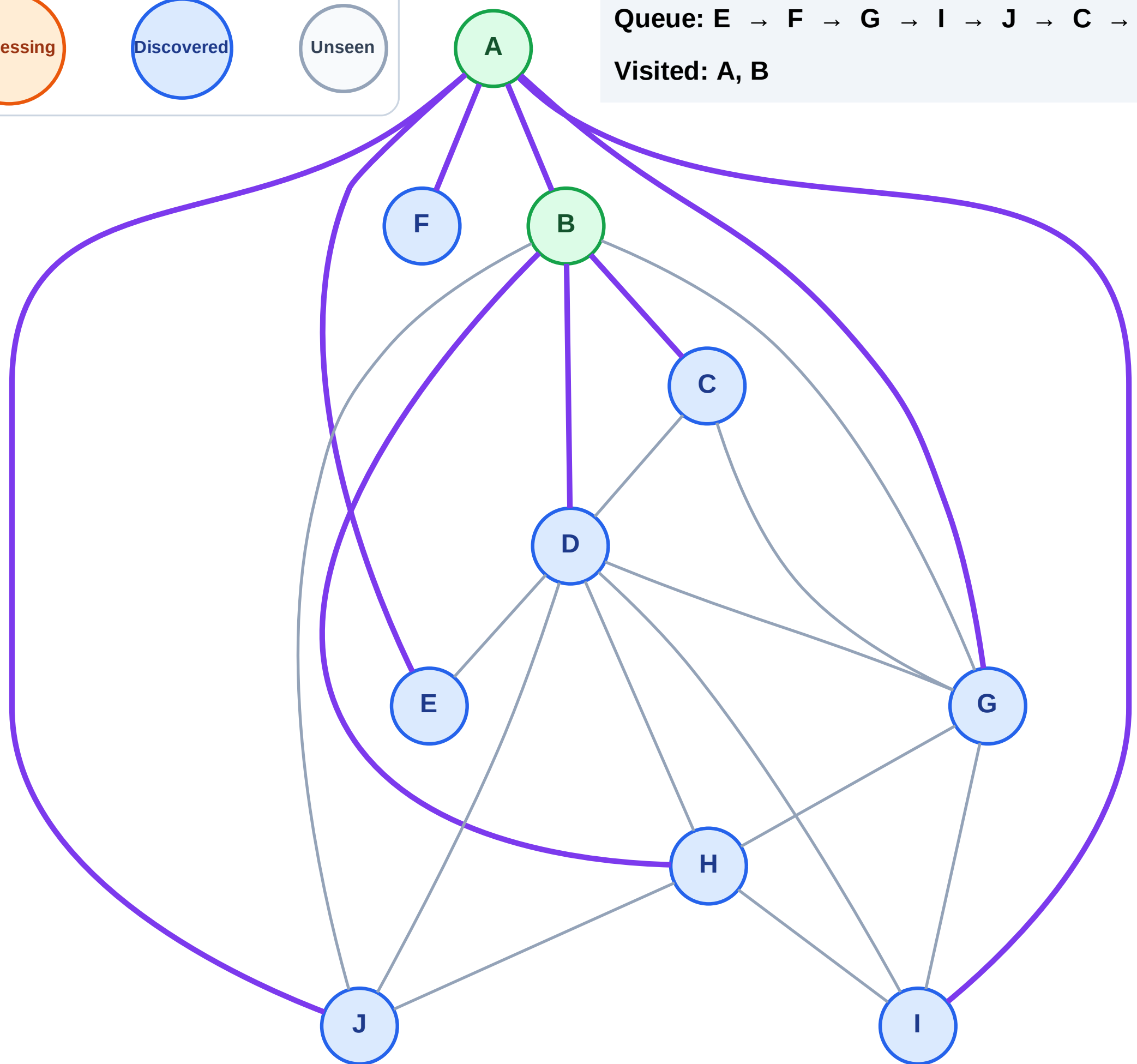
Step 5: Discover C, D, H from B

Legend



Queue: E → F → G → I → J → C → D → H

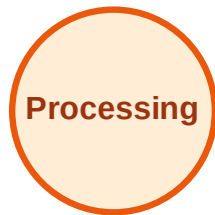
Visited: A, B



BFS Traversal

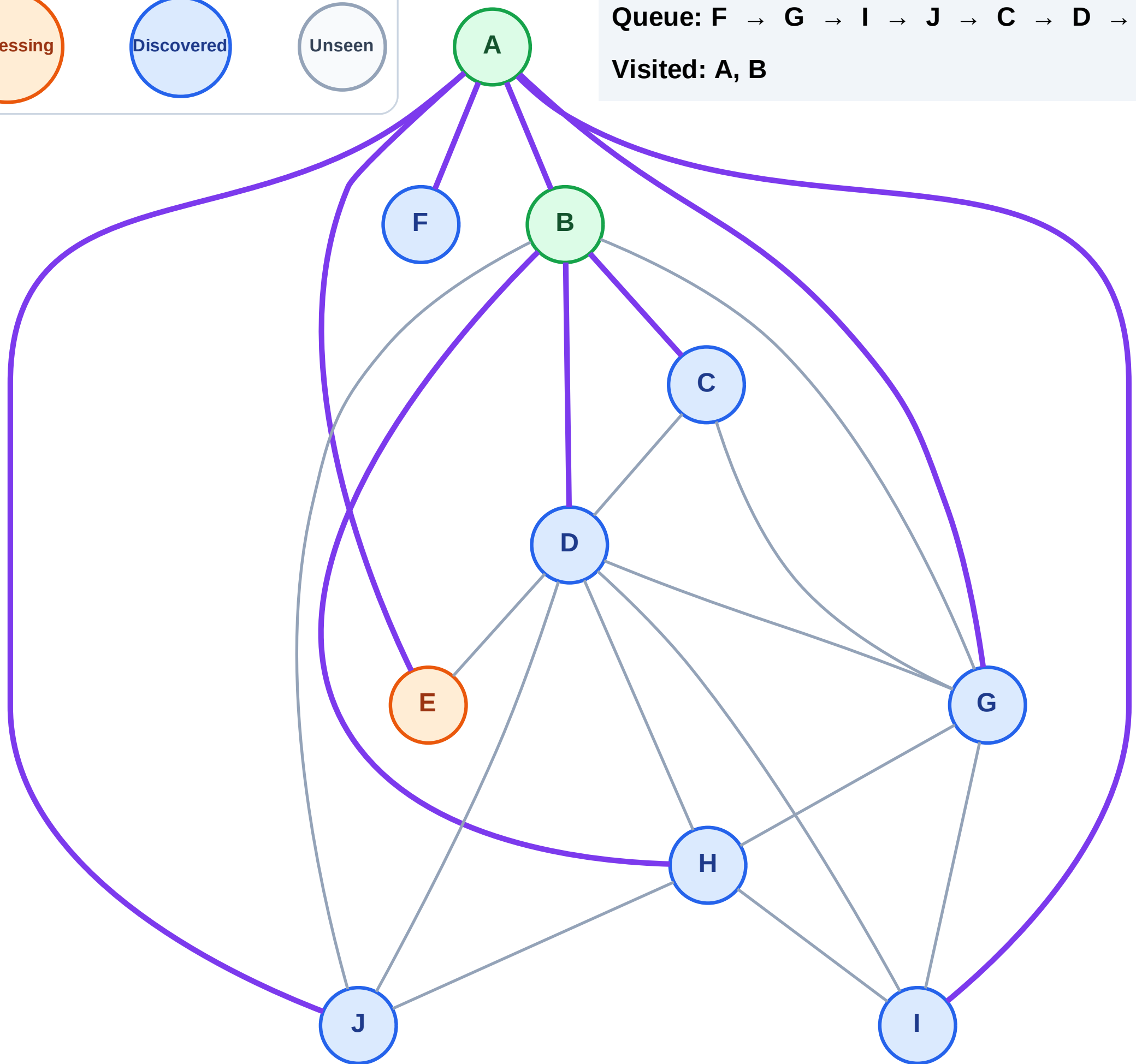
Step 6: Process node E

Legend



Queue: F → G → I → J → C → D → H

Visited: A, B

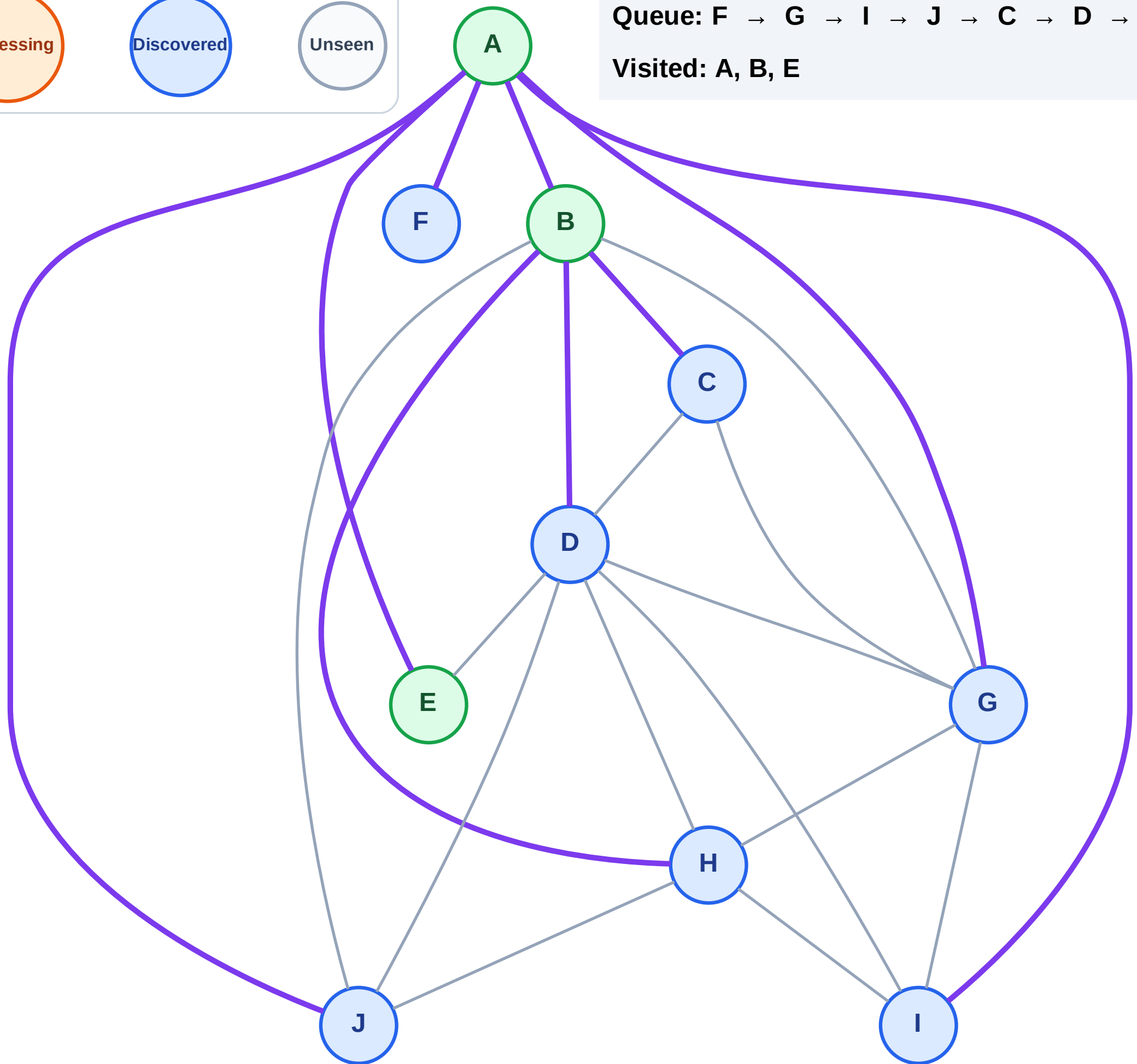


Step 7: No new neighbors from E

Legend

Unseen

Visited: A, B, E



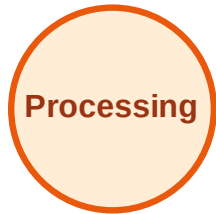
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



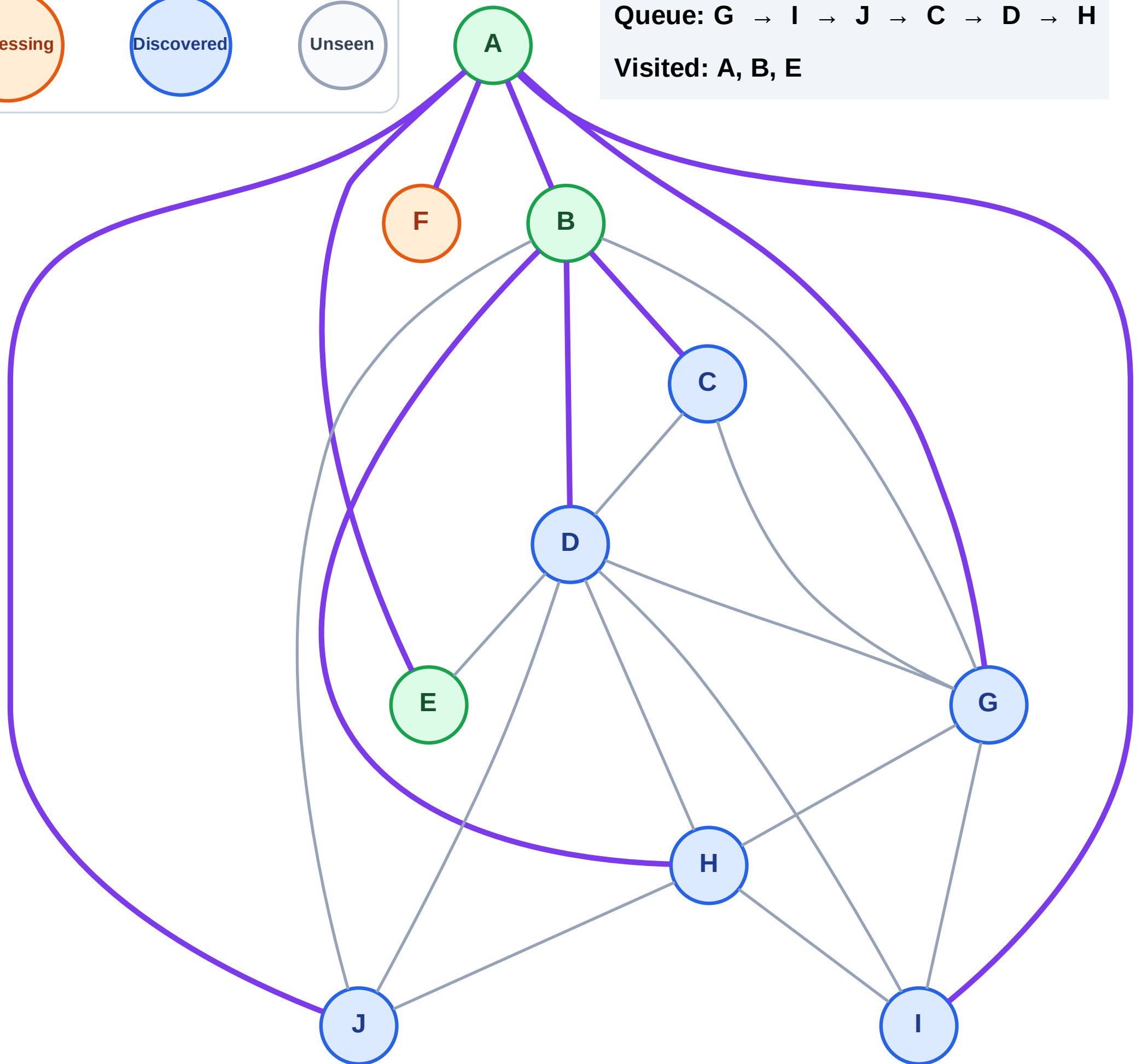
Discovered



Unseen

Queue: G → I → J → C → D → H

Visited: A, B, E



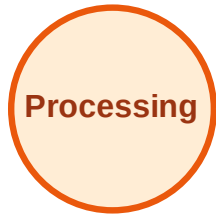
BFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



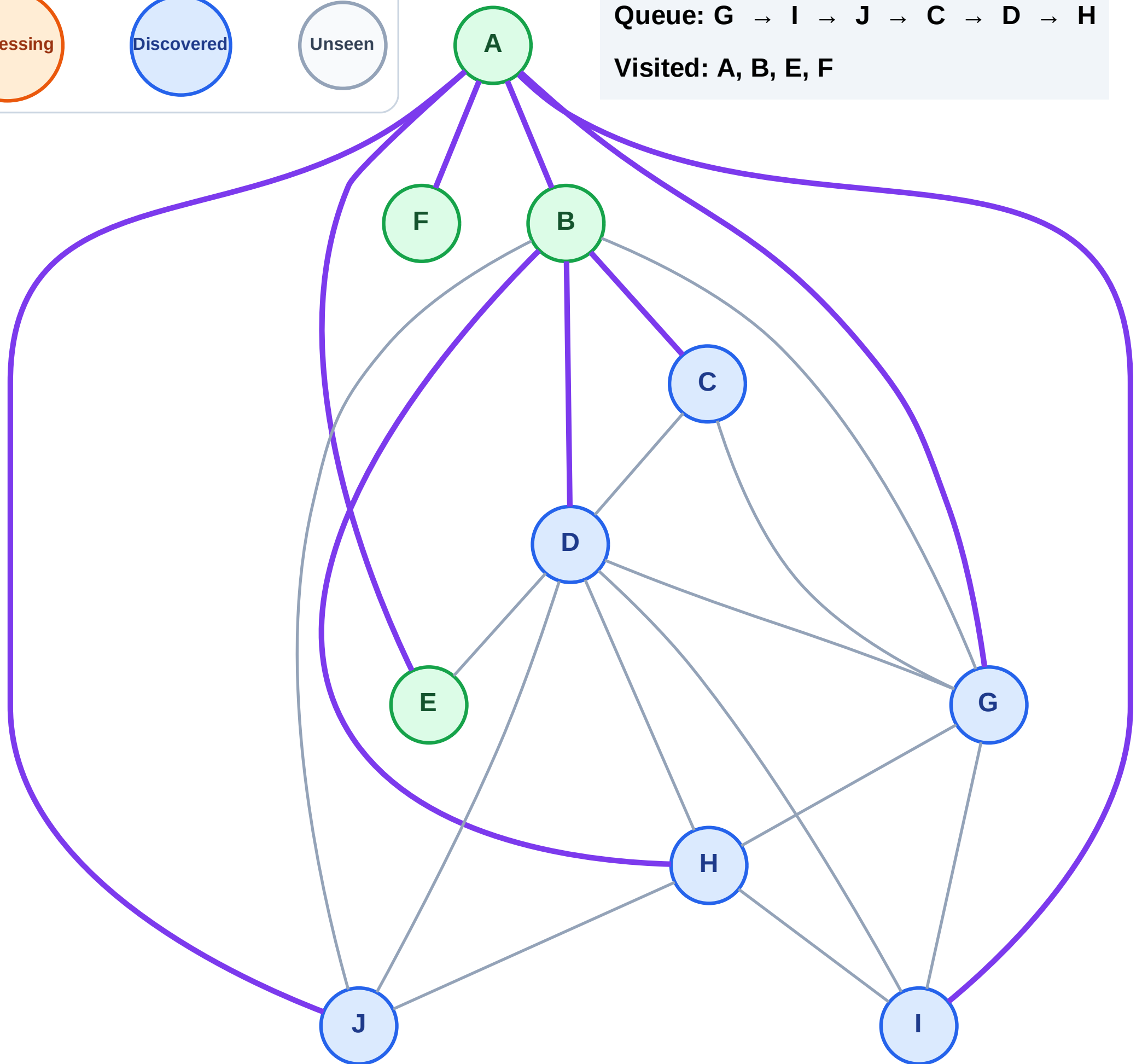
Discovered



Unseen

Queue: G → I → J → C → D → H

Visited: A, B, E, F



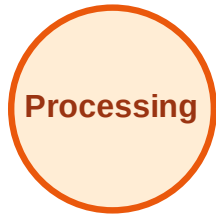
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



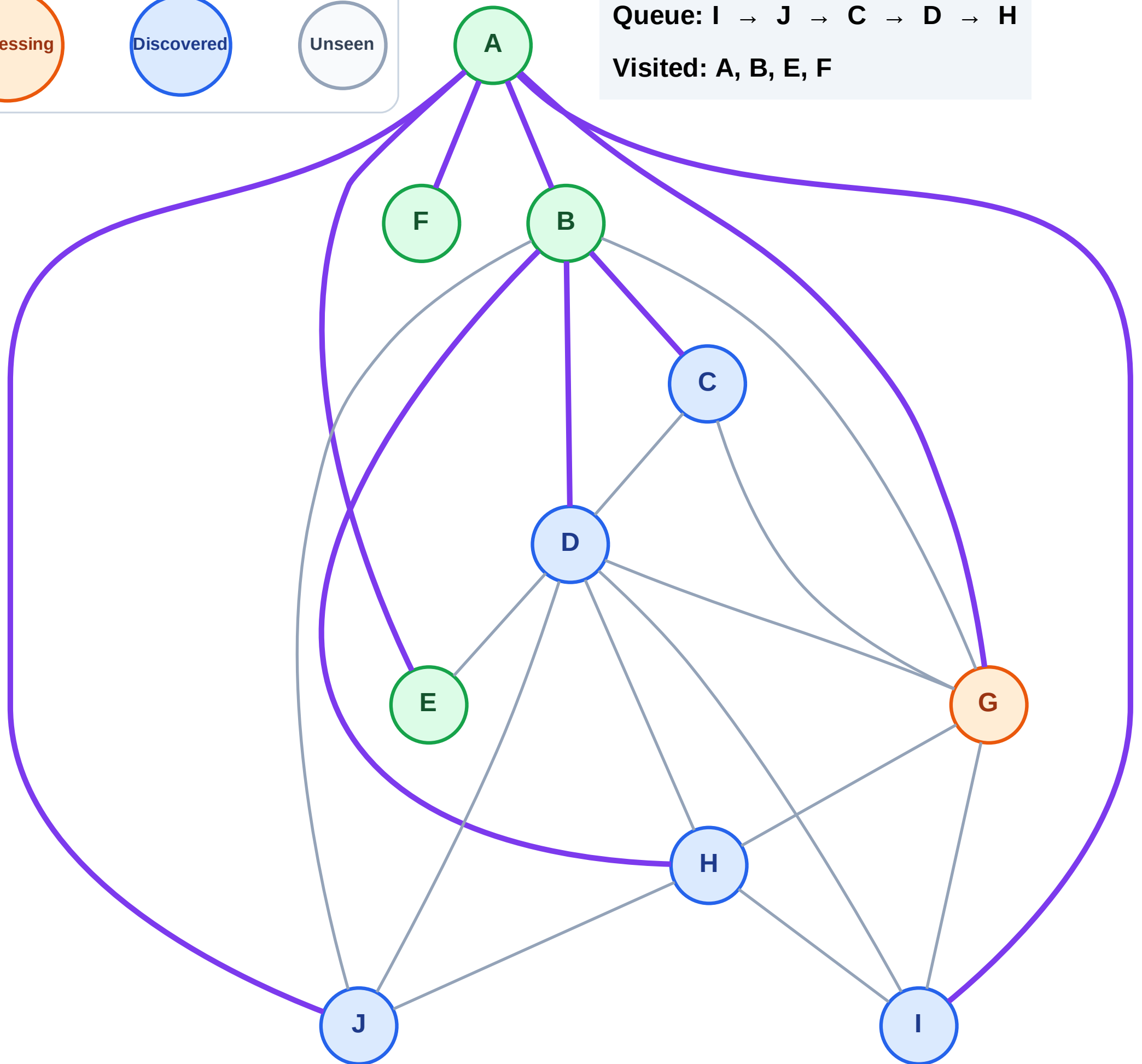
Discovered



Unseen

Queue: I → J → C → D → H

Visited: A, B, E, F



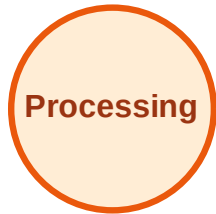
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



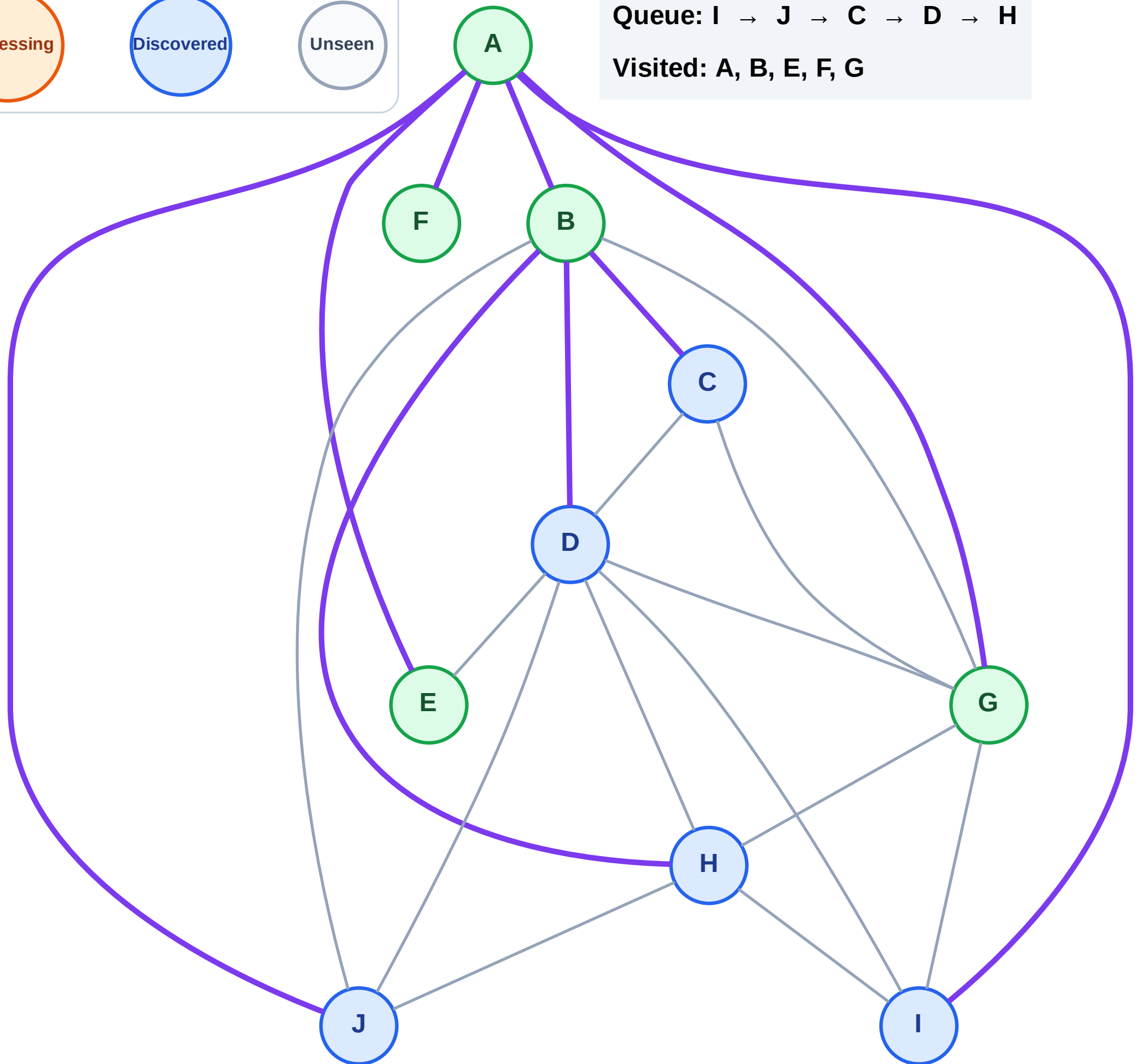
Discovered



Unseen

Queue: I → J → C → D → H

Visited: A, B, E, F, G



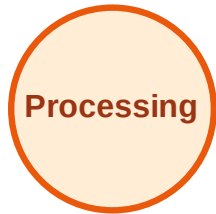
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



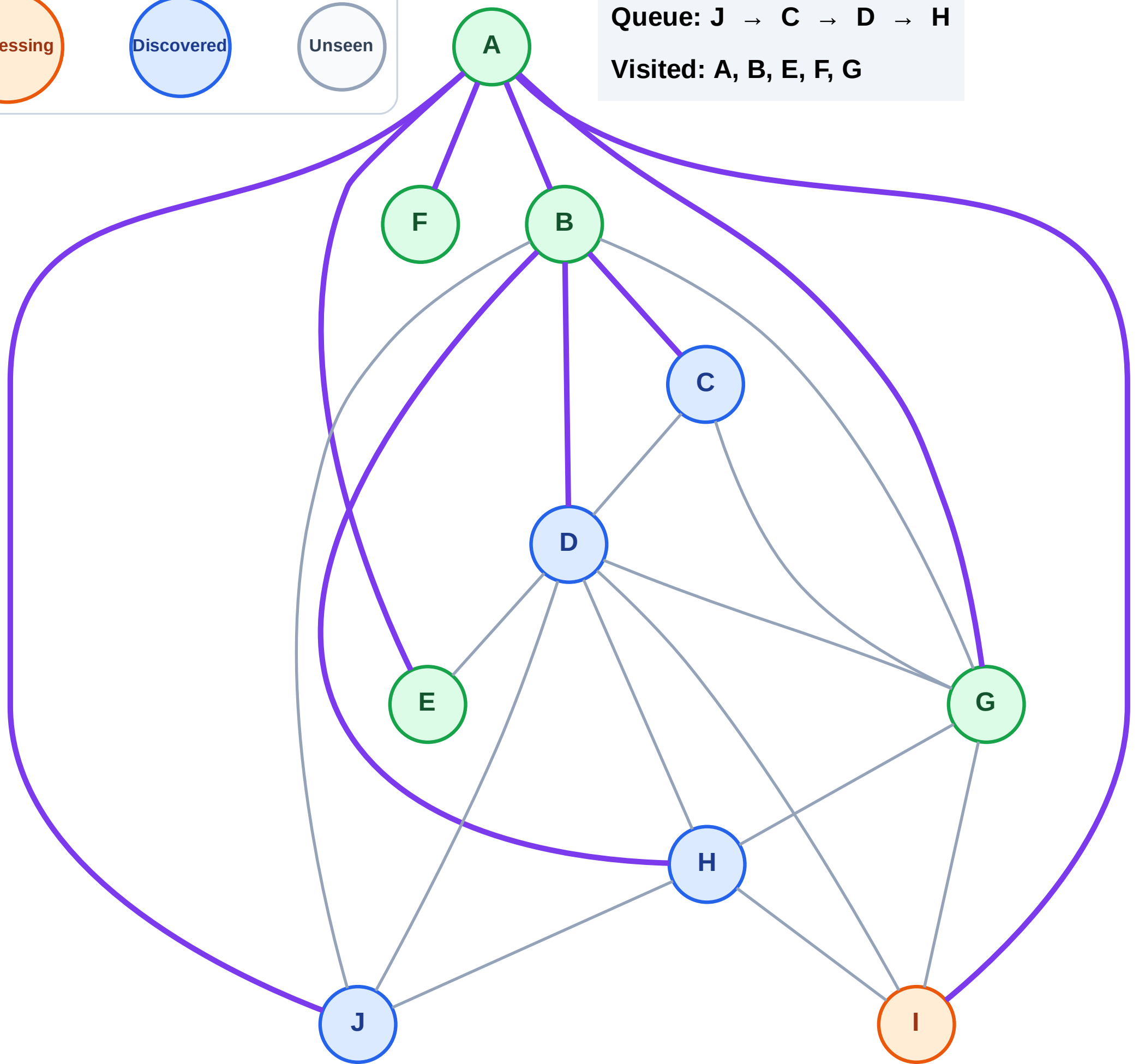
Discovered



Unseen

Queue: J → C → D → H

Visited: A, B, E, F, G



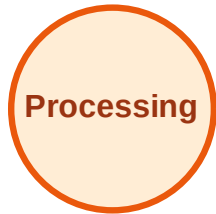
BFS Traversal

Step 13: No new neighbors from I

Legend



Complete



Processing



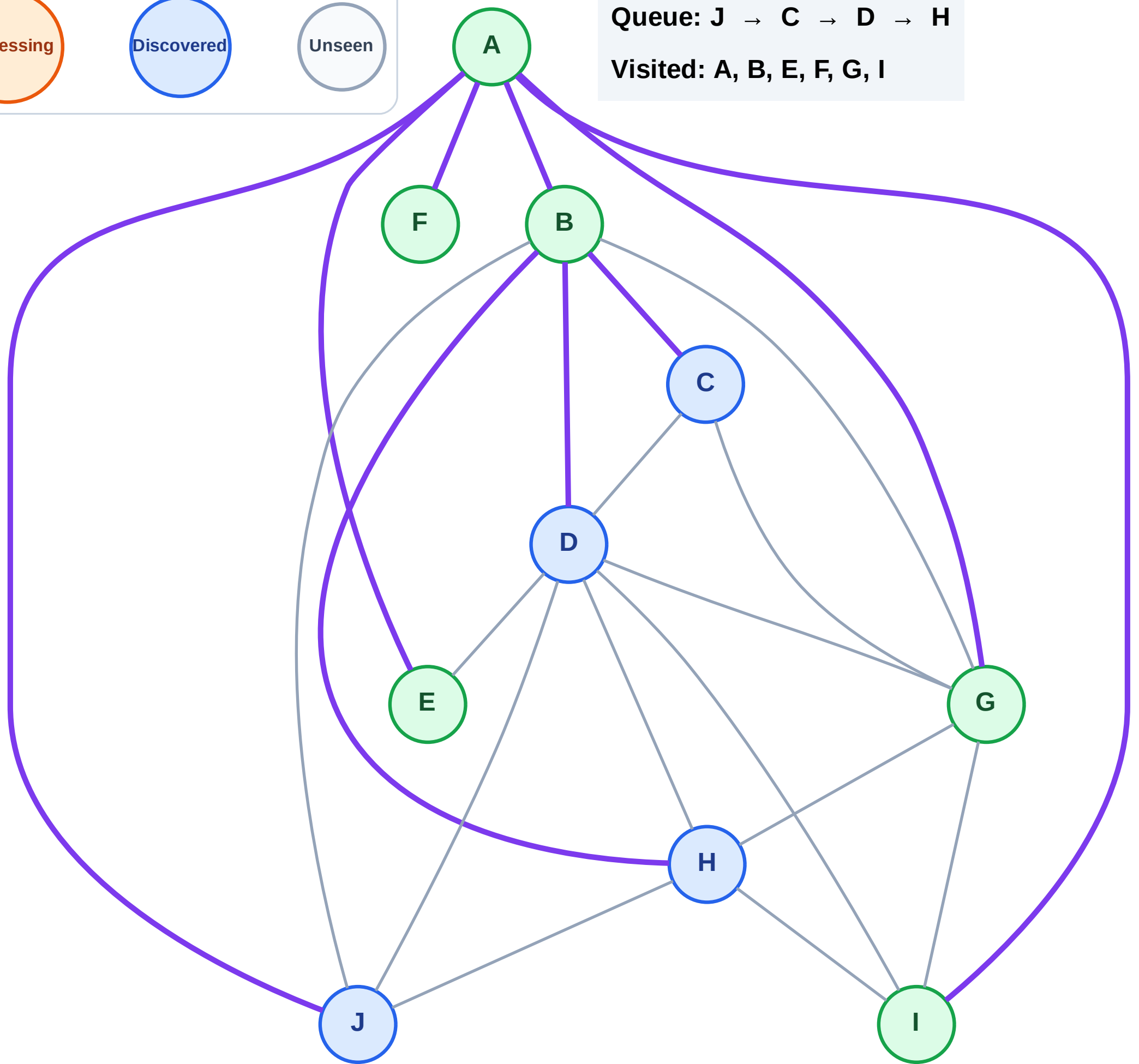
Discovered



Unseen

Queue: J → C → D → H

Visited: A, B, E, F, G, I



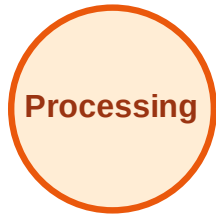
BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



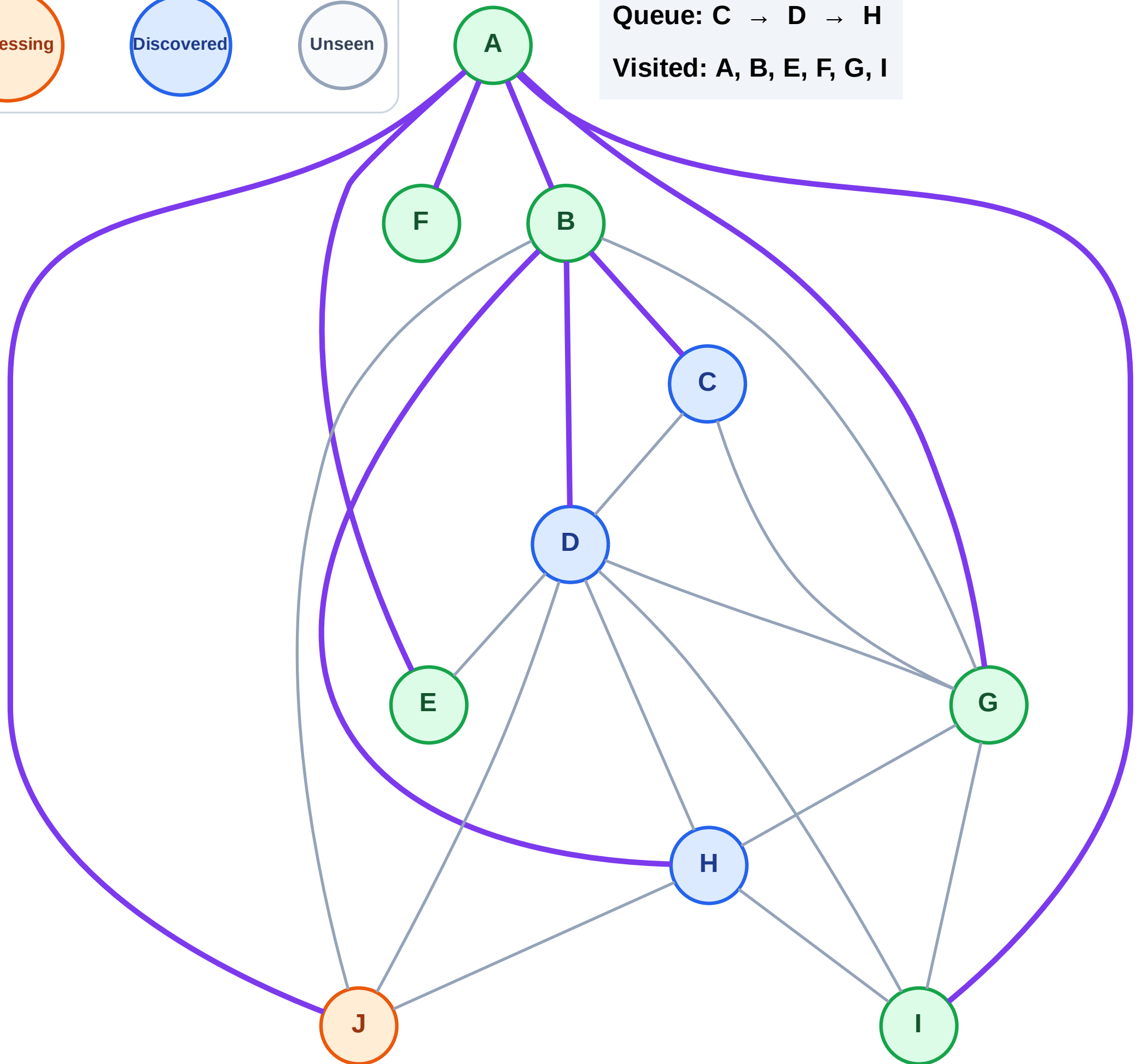
Discovered



Unseen

Queue: C → D → H

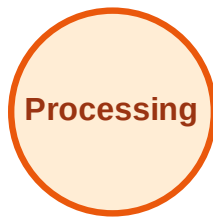
Visited: A, B, E, F, G, I



BFS Traversal

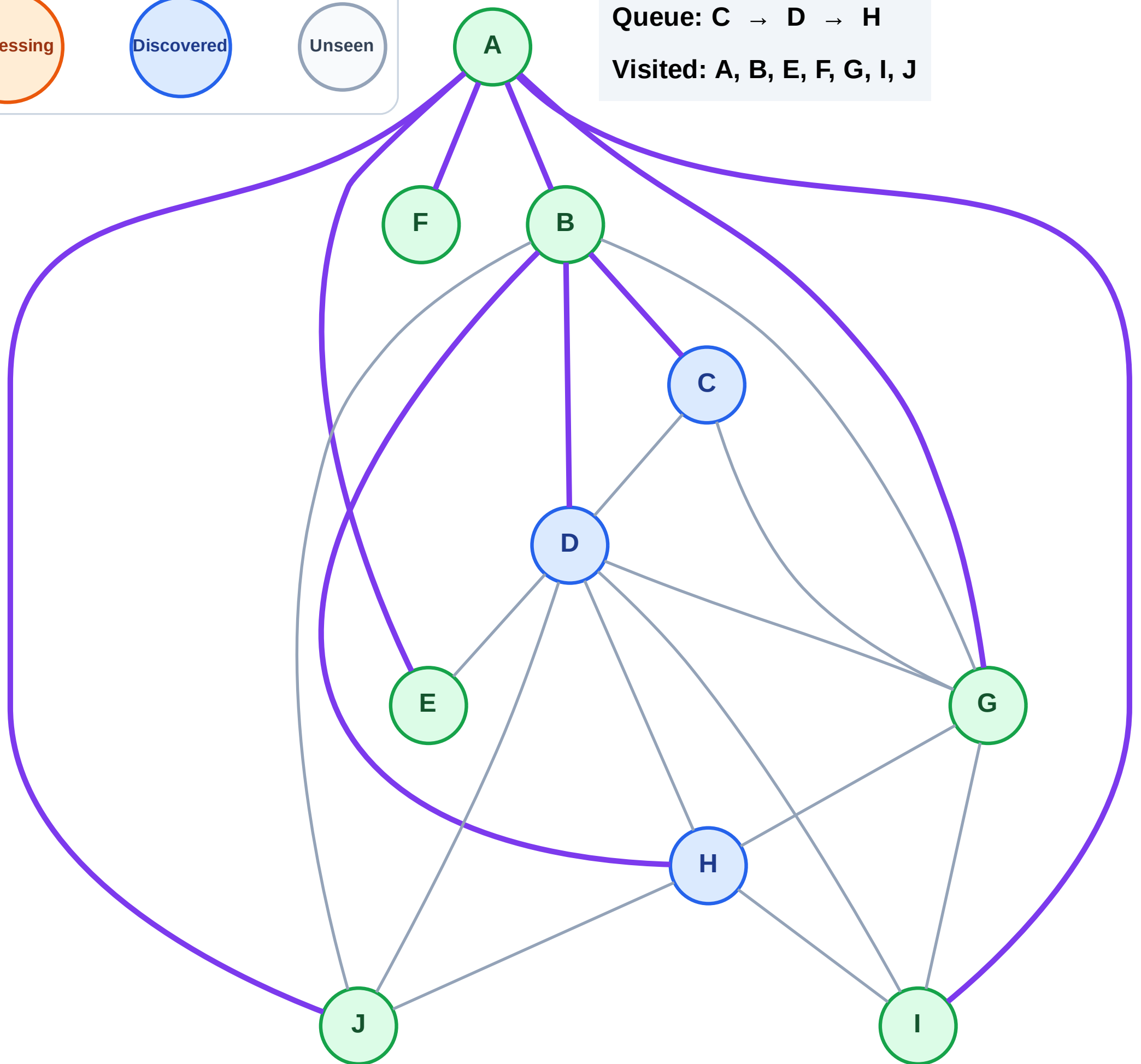
Step 15: No new neighbors from J

Legend

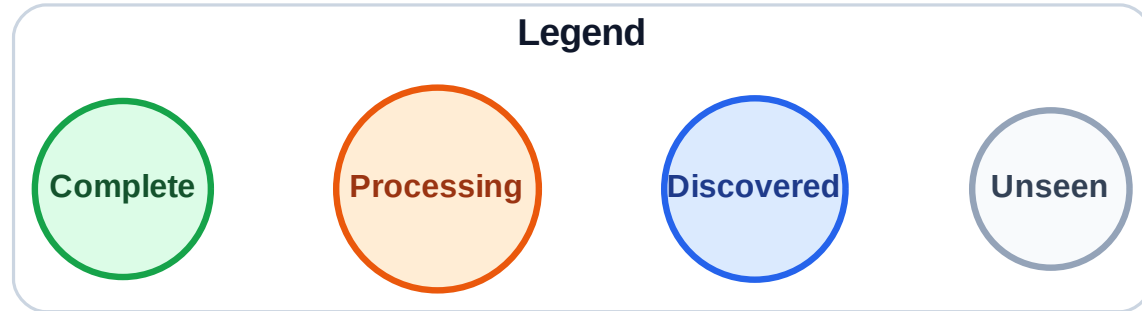


Queue: C → D → H

Visited: A, B, E, F, G, I, J

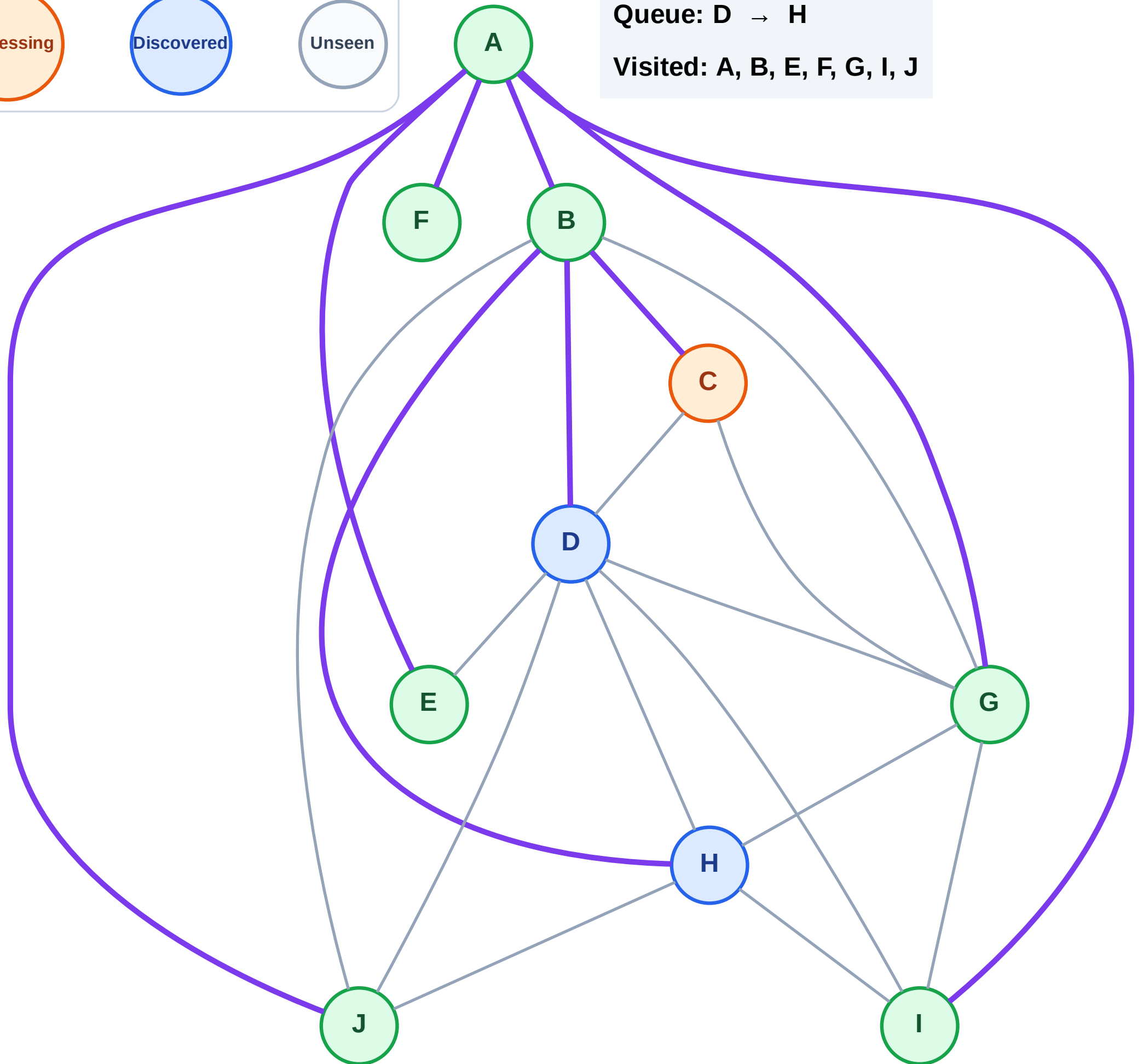


Step 16: Process node C



Queue: D → H

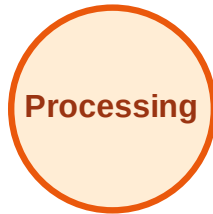
Visited: A, B, E, F, G, I, J



BFS Traversal

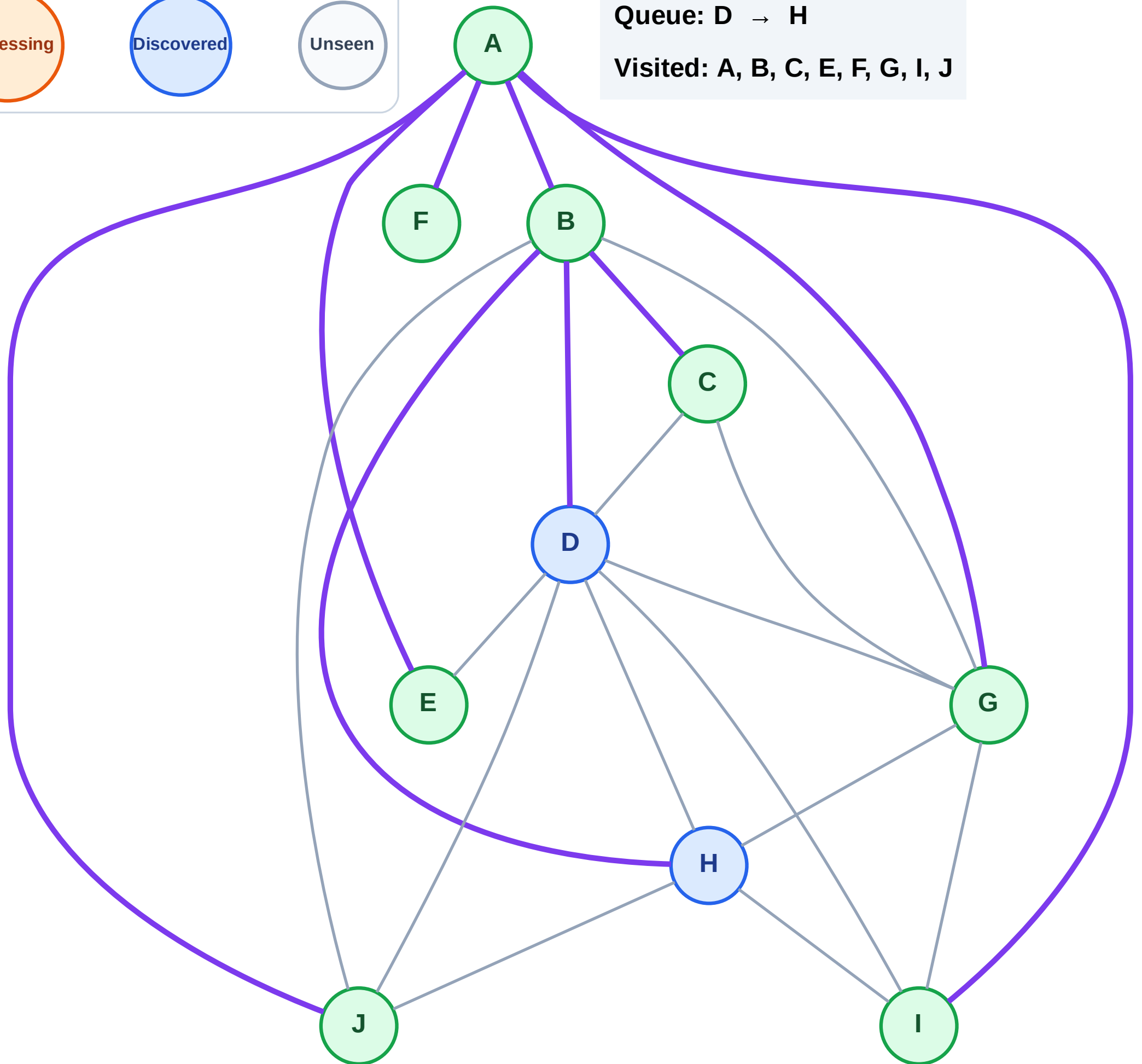
Step 17: No new neighbors from C

Legend



Queue: D → H

Visited: A, B, C, E, F, G, I, J



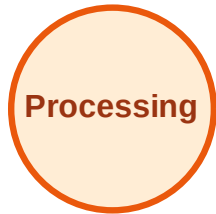
BFS Traversal

Step 18: Process node D

Legend



Complete



Processing



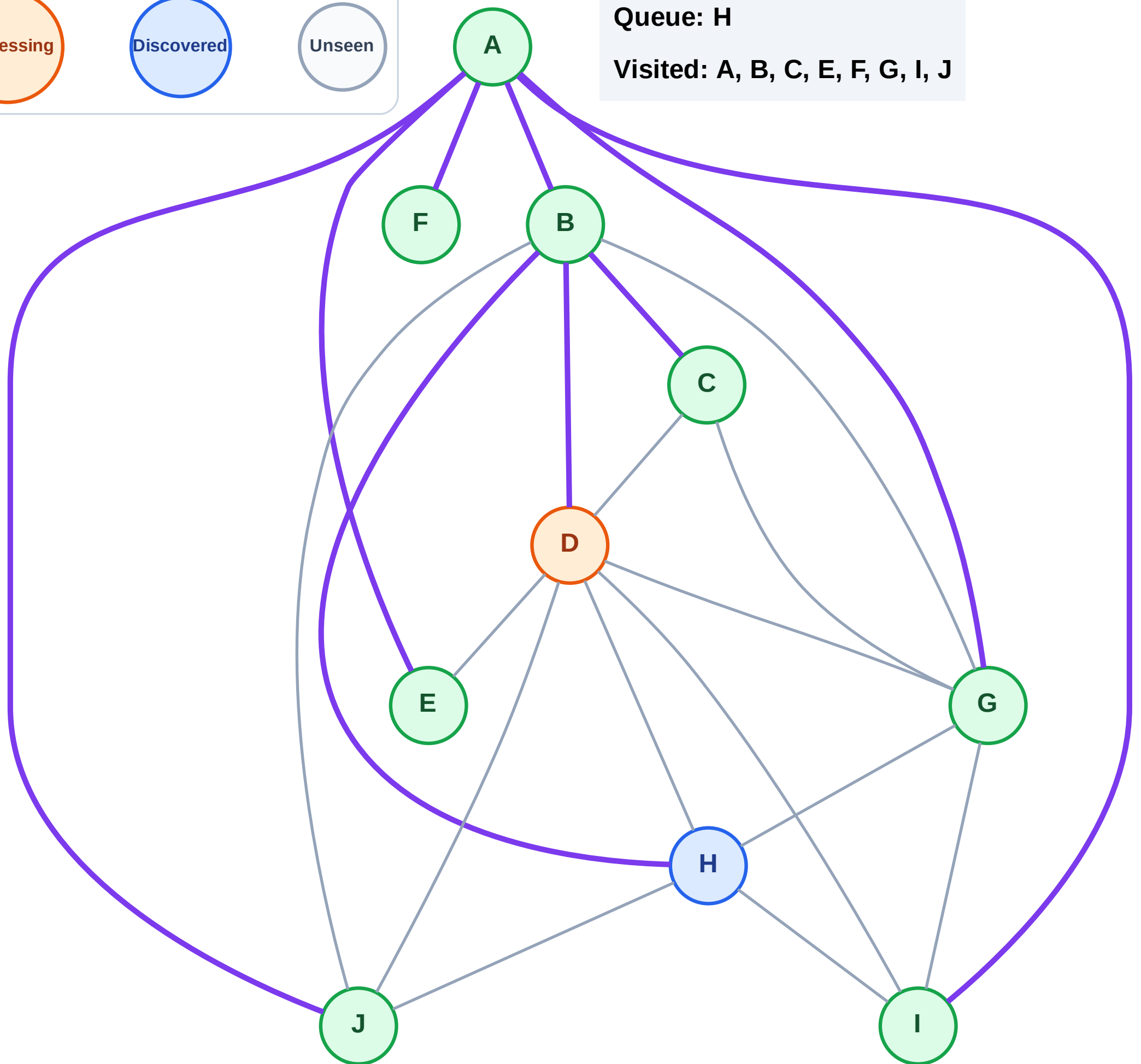
Discovered



Unseen

Queue: H

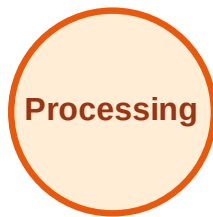
Visited: A, B, C, E, F, G, I, J



BFS Traversal

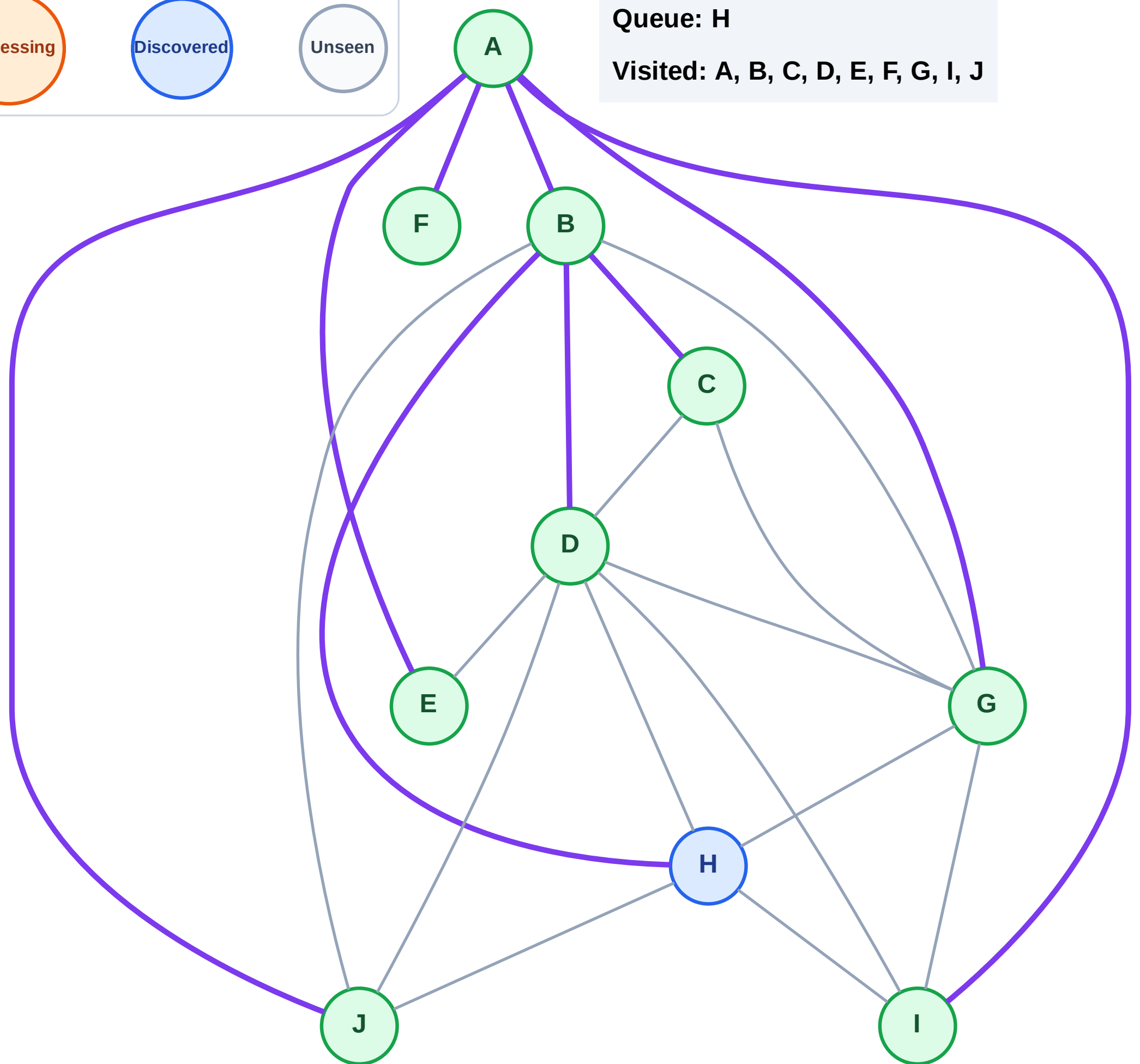
Step 19: No new neighbors from D

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



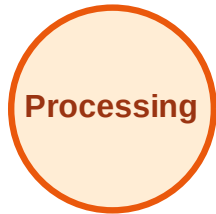
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



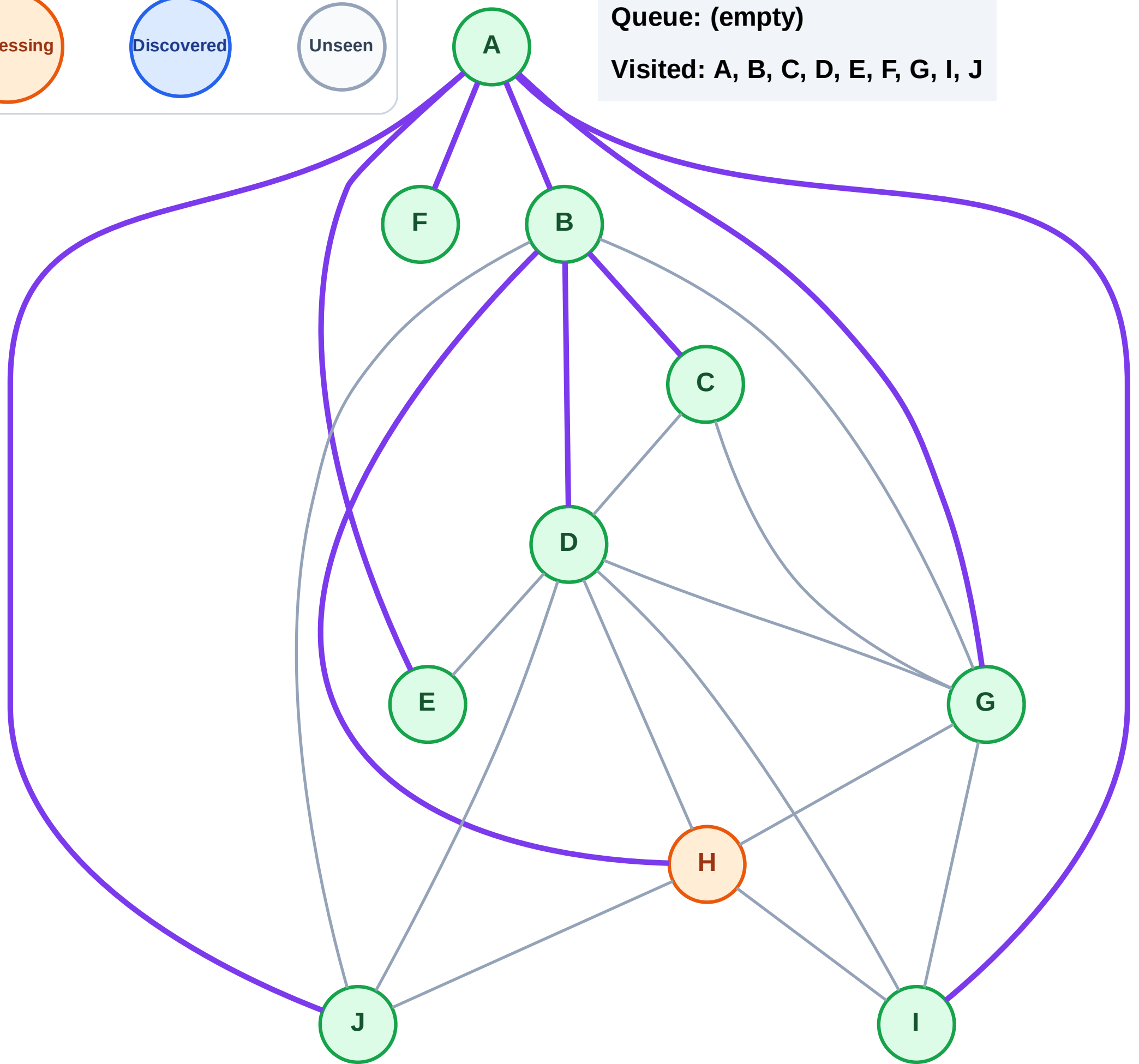
Discovered



Unseen

Queue: (empty)

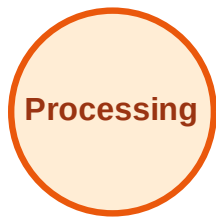
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

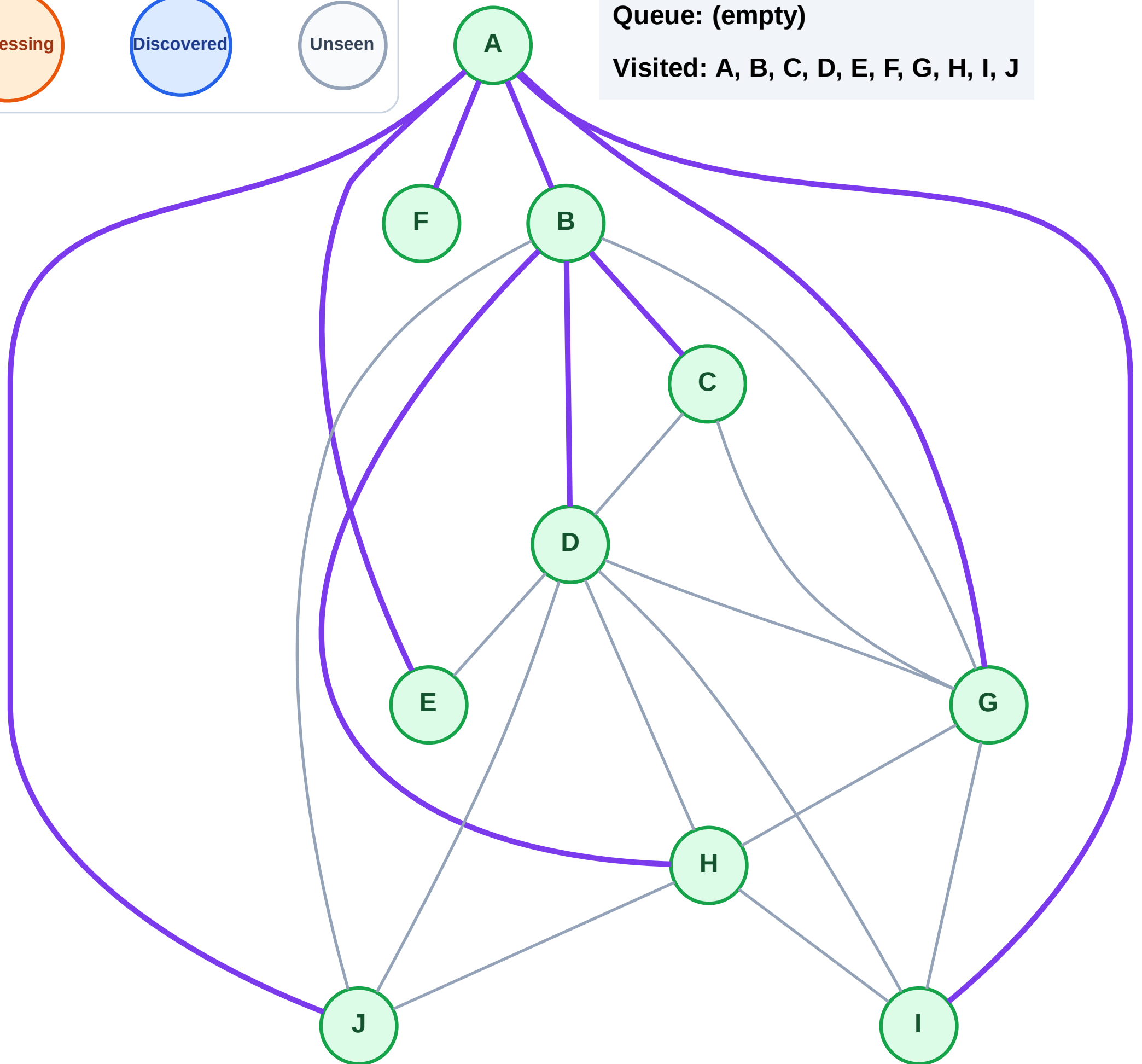
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



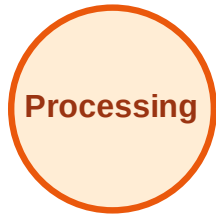
DFS Traversal

Step 1: Start at A

Legend



Complete



Processing



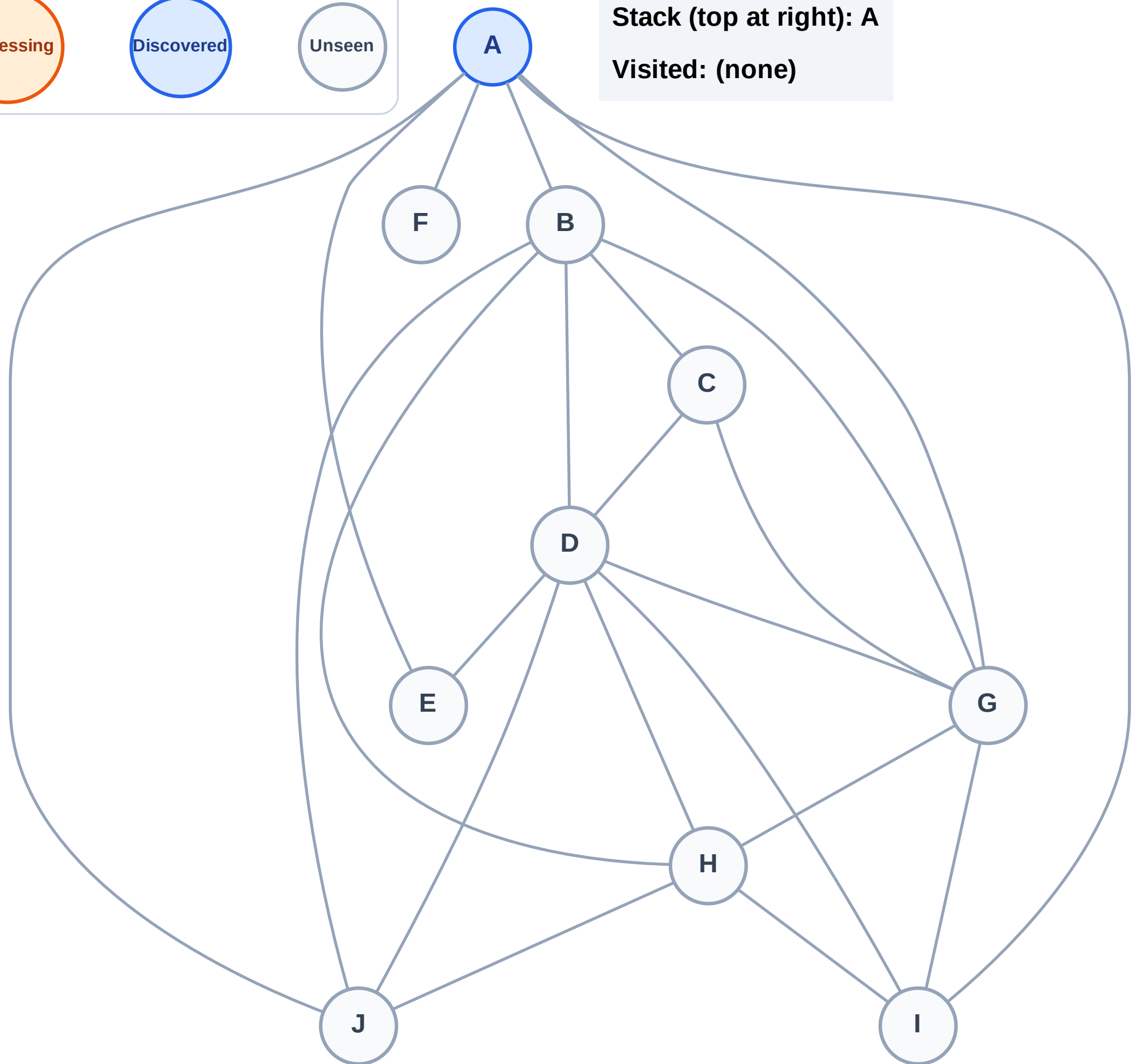
Discovered



Unseen

Stack (top at right): A

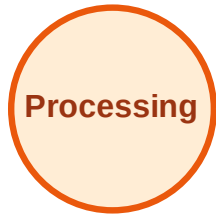
Visited: (none)



DFS Traversal

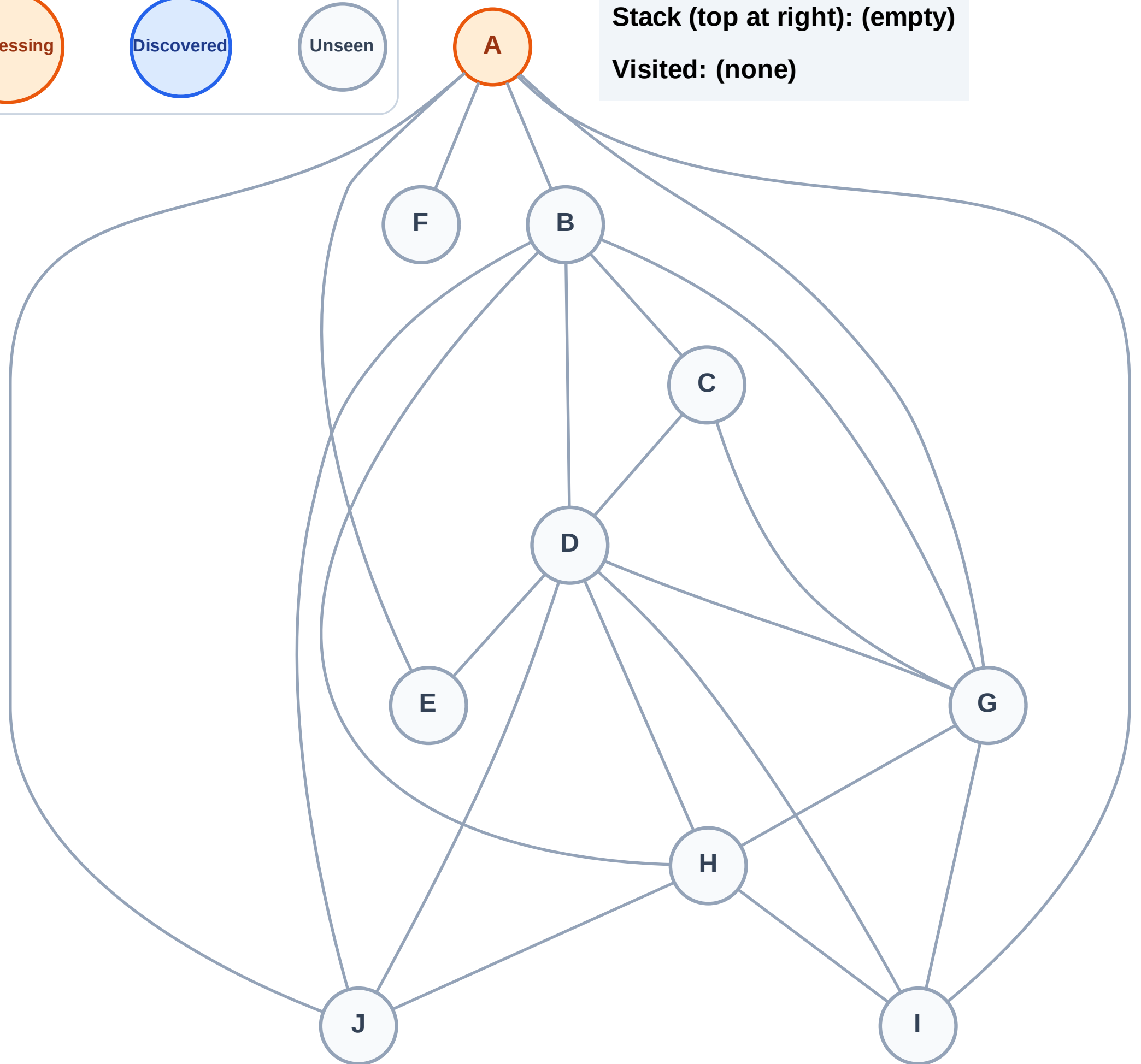
Step 2: Process node A

Legend



Stack (top at right): (empty)

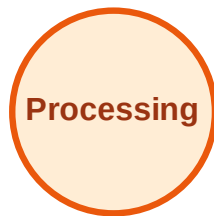
Visited: (none)



DFS Traversal

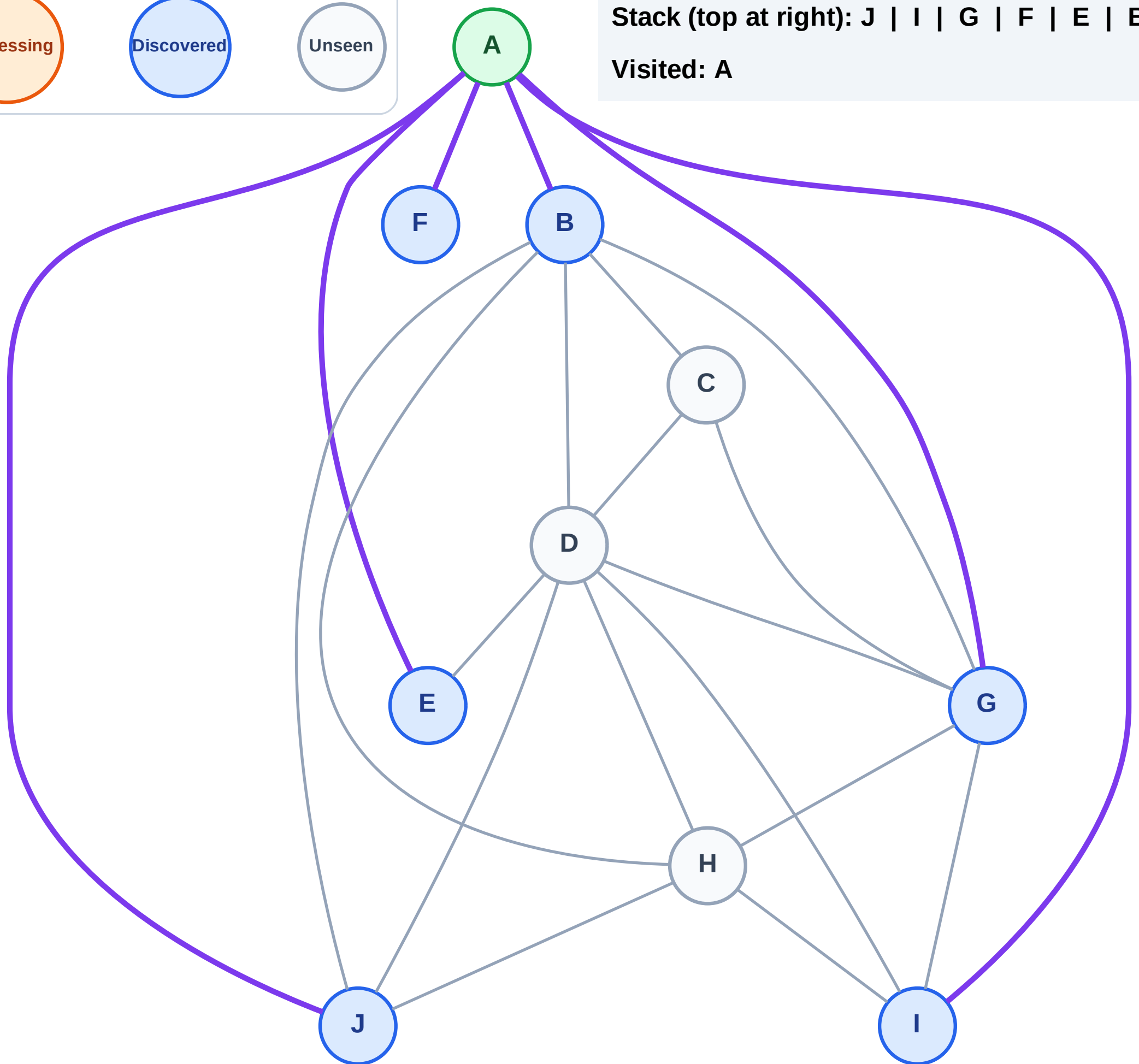
Step 3: Discover J, I, G, F, E, B from A

Legend



Stack (top at right): J | I | G | F | E | B

Visited: A



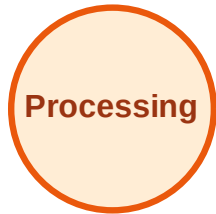
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



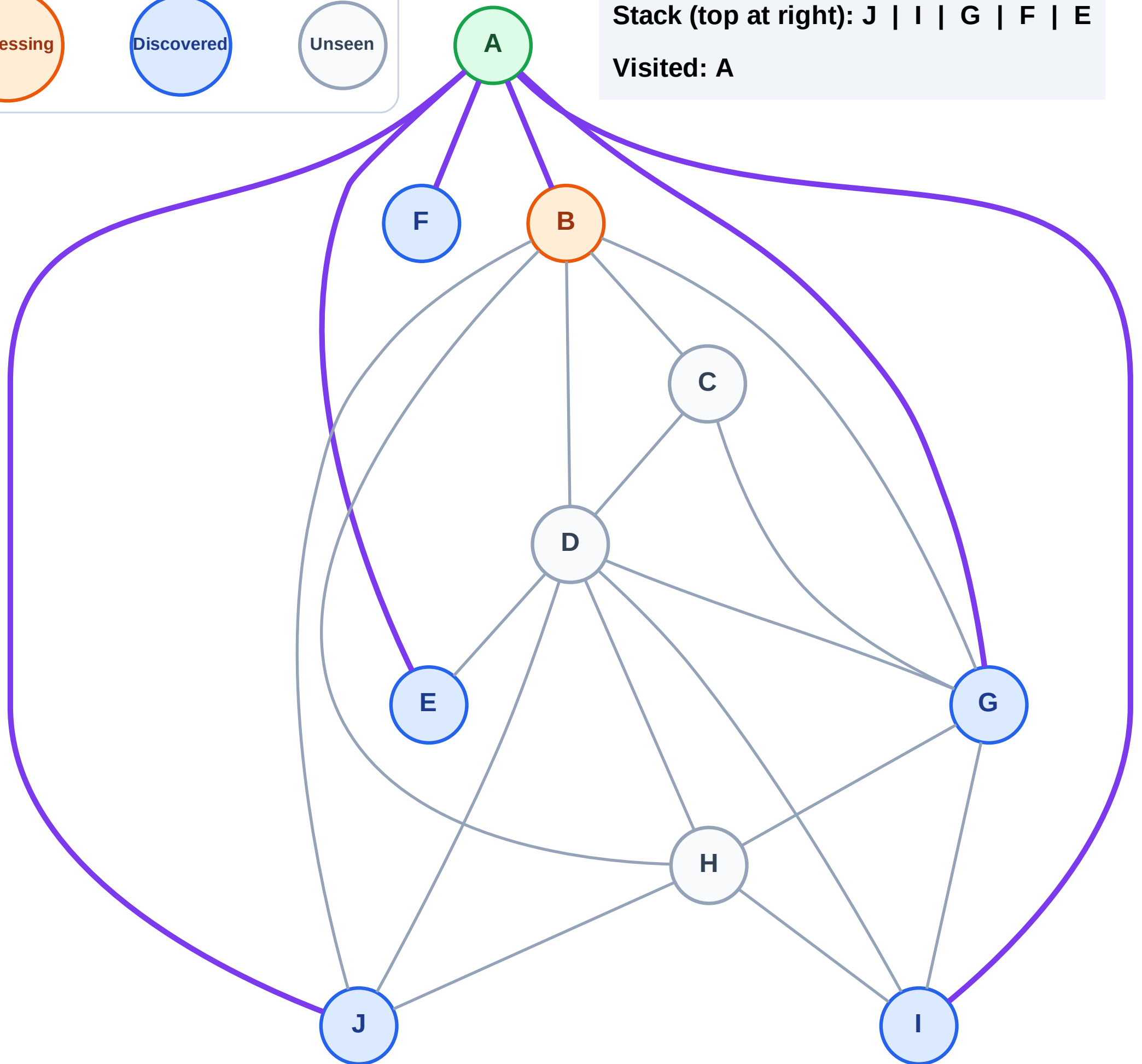
Discovered



Unseen

Stack (top at right): J | I | G | F | E

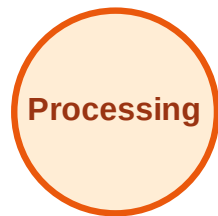
Visited: A



DFS Traversal

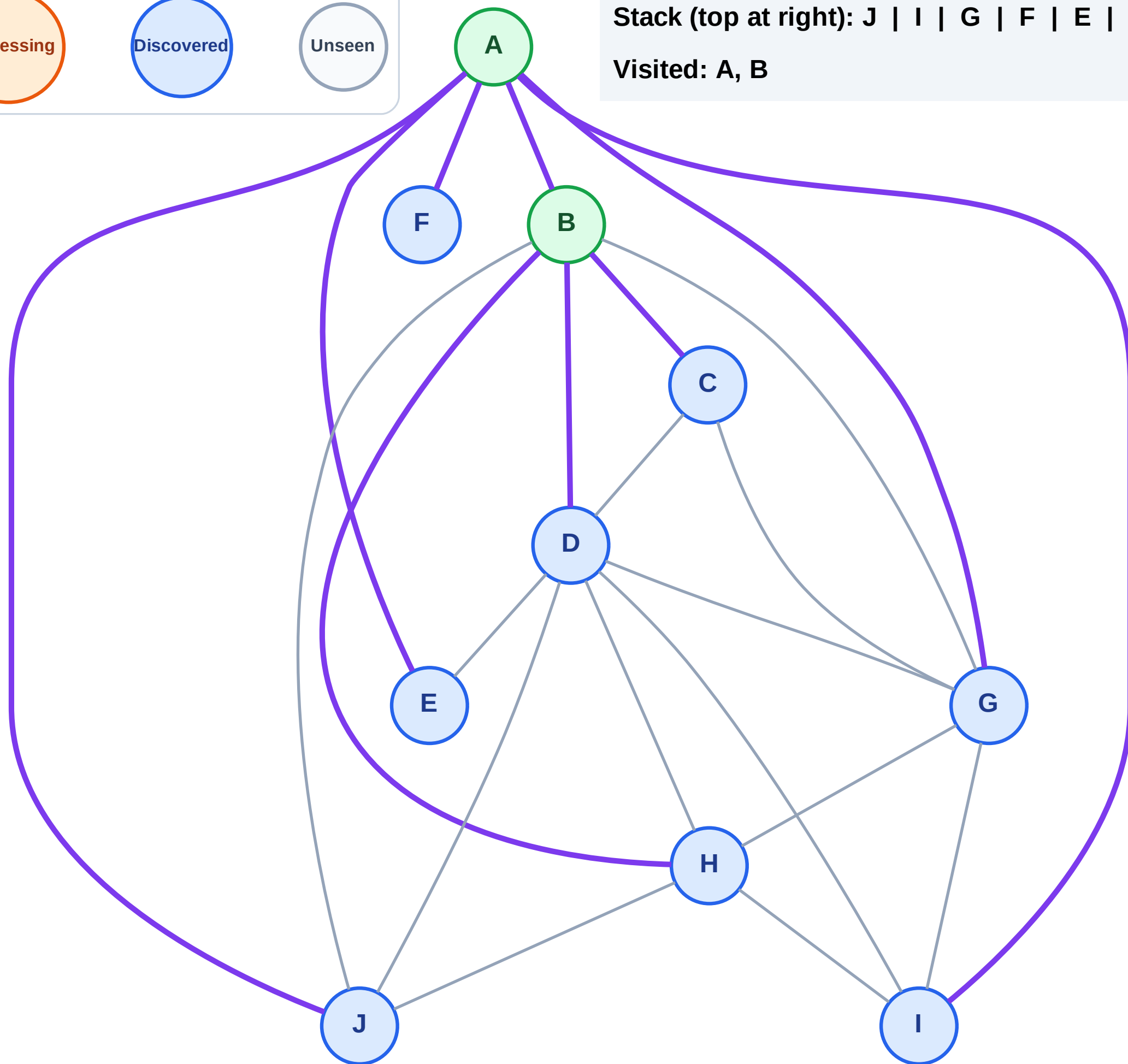
Step 5: Discover H, D, C from B

Legend



Stack (top at right): J | I | G | F | E | H | D | C

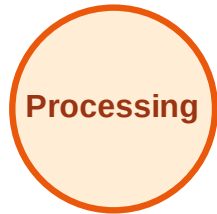
Visited: A, B



DFS Traversal

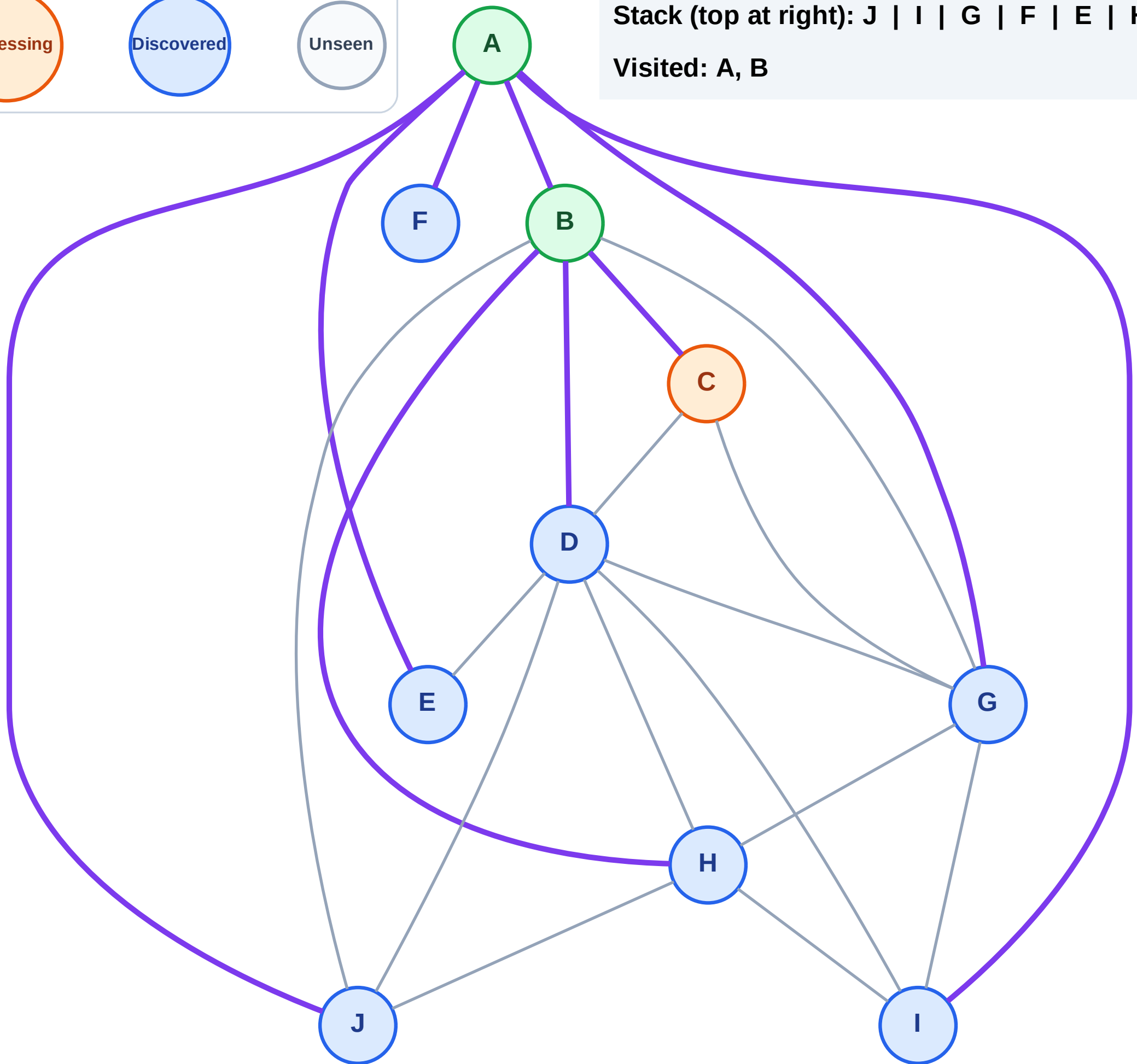
Step 6: Process node C

Legend



Stack (top at right): J | I | G | F | E | H | D

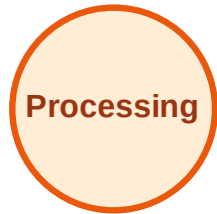
Visited: A, B



DFS Traversal

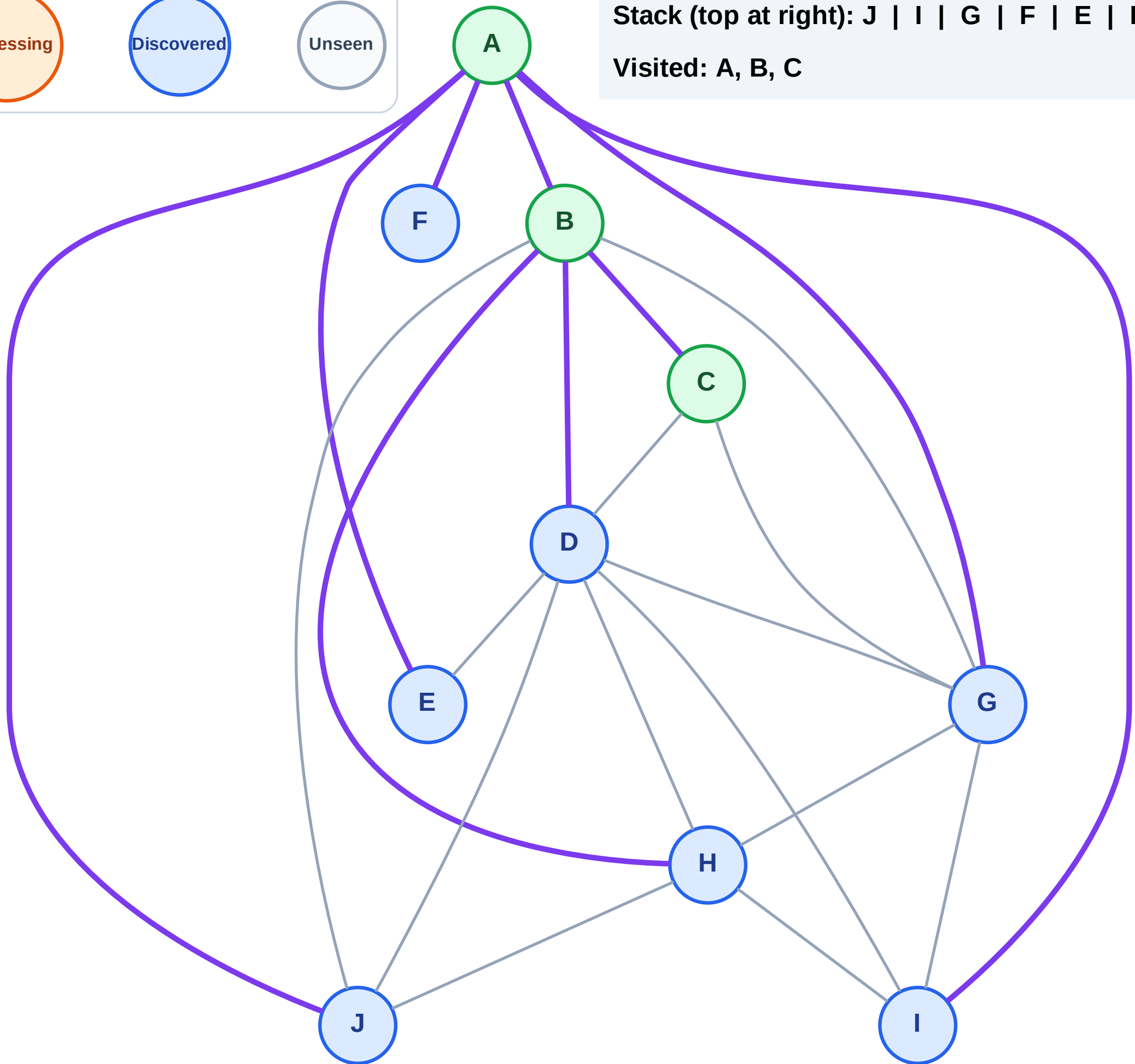
Step 7: No new neighbors from C

Legend



Stack (top at right): J | I | G | F | E | H | D

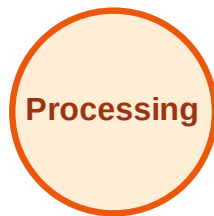
Visited: A, B, C



DFS Traversal

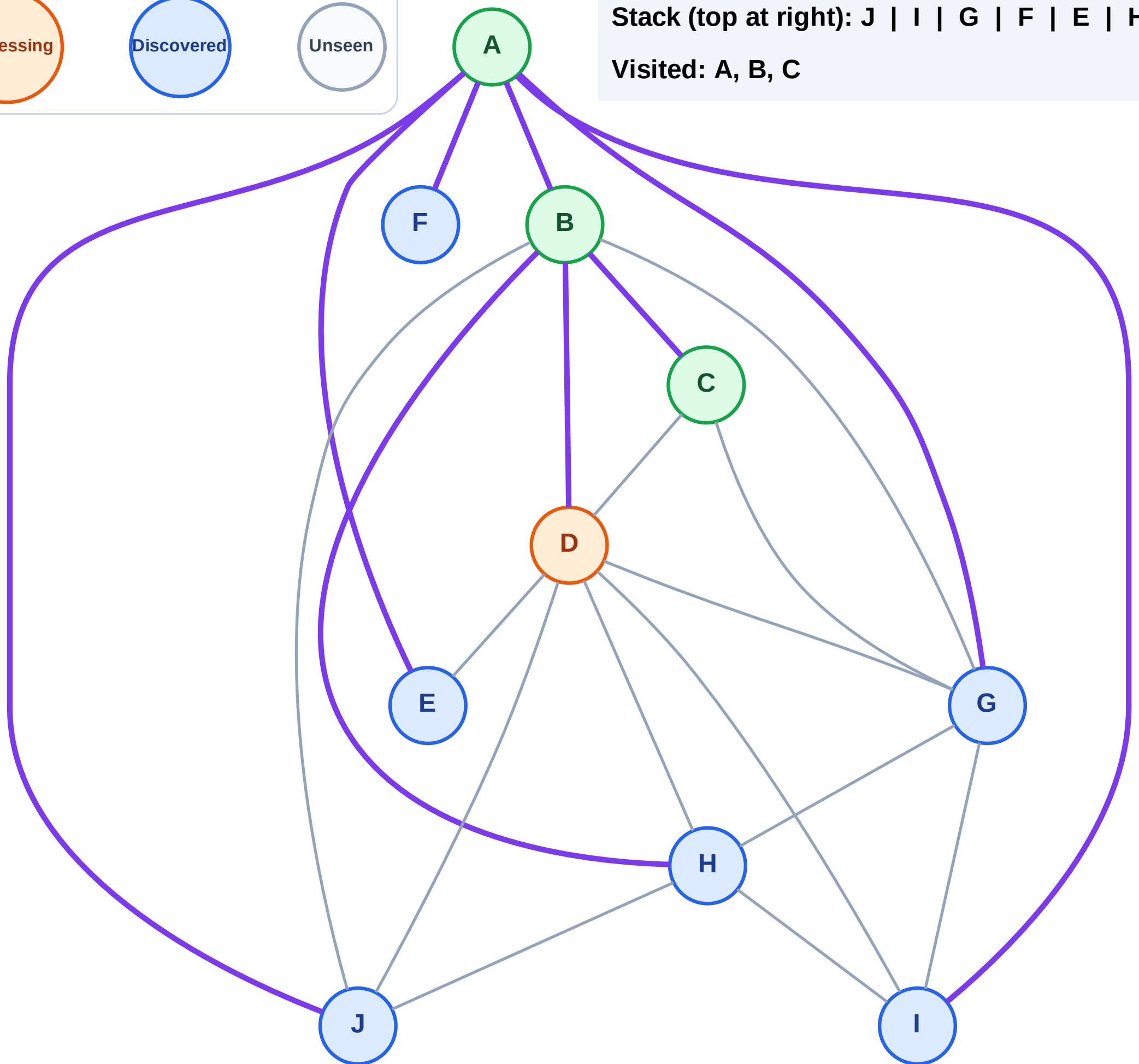
Step 8: Process node D

Legend



Stack (top at right): J | I | G | F | E | H

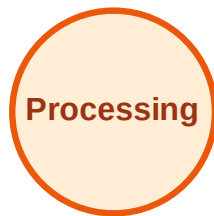
Visited: A, B, C



DFS Traversal

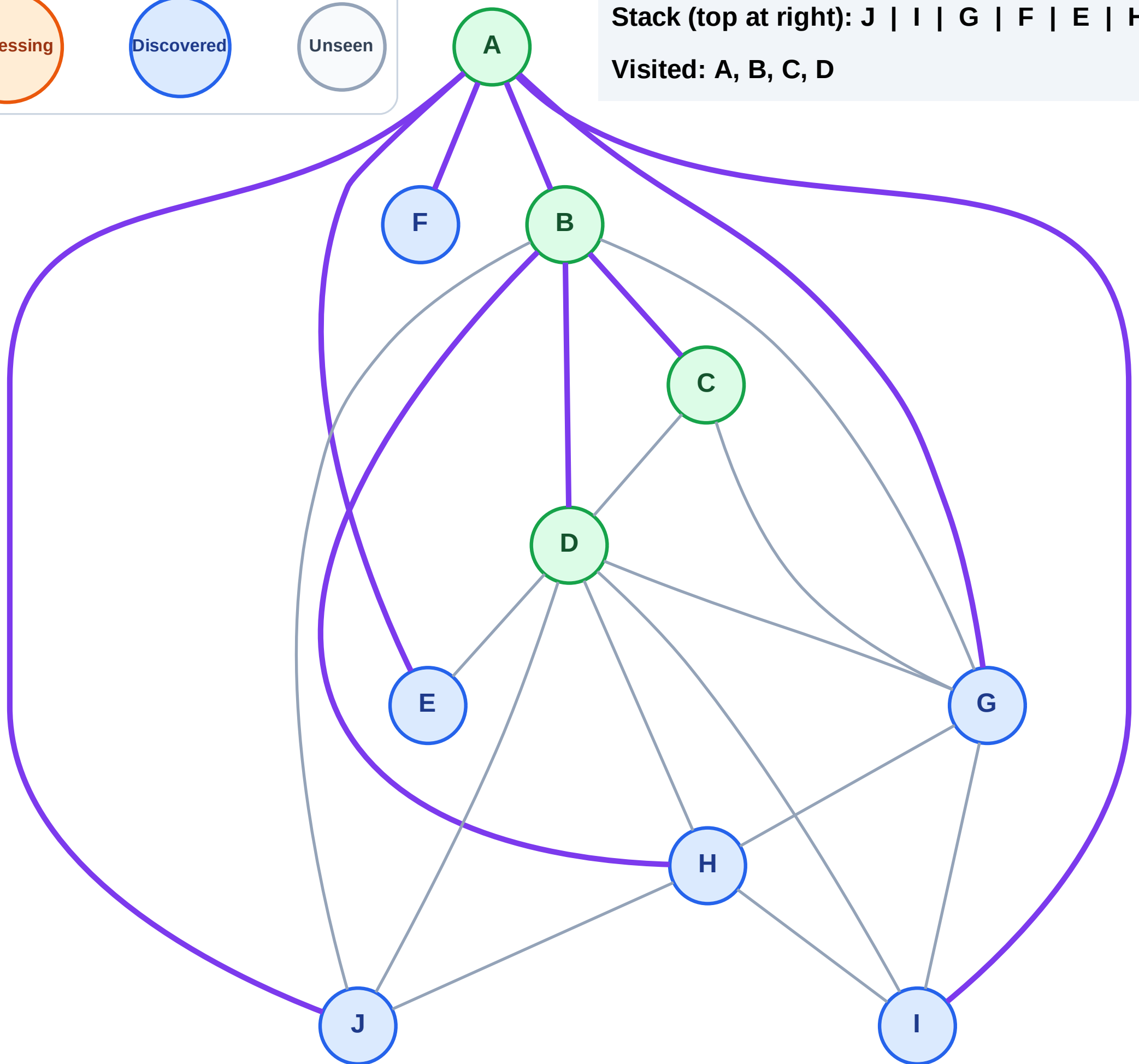
Step 9: No new neighbors from D

Legend



Stack (top at right): J | I | G | F | E | H

Visited: A, B, C, D



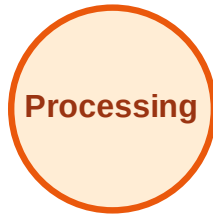
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



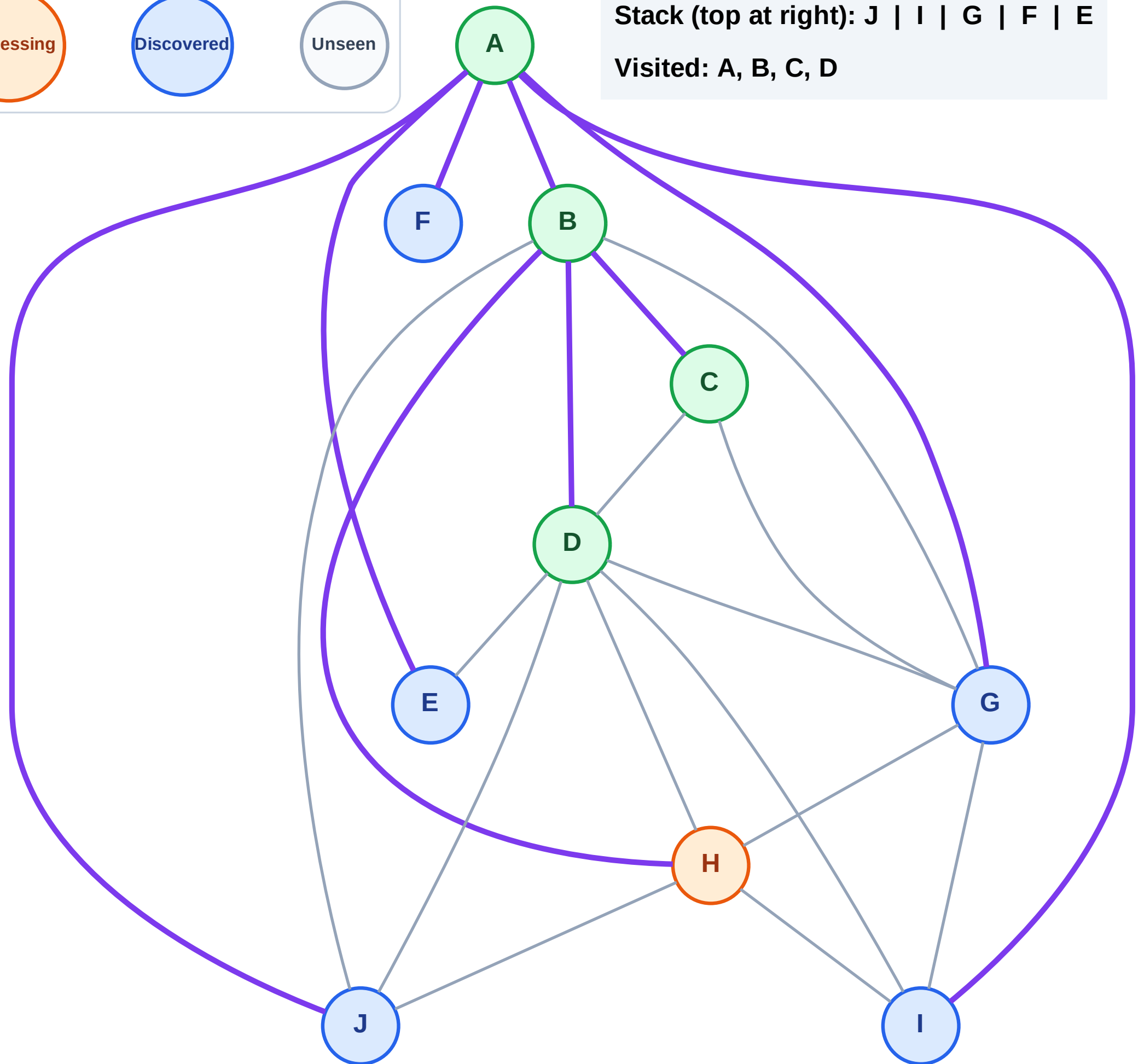
Discovered



Unseen

Stack (top at right): J | I | G | F | E

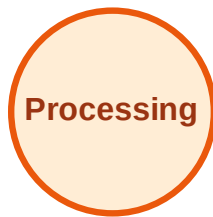
Visited: A, B, C, D



DFS Traversal

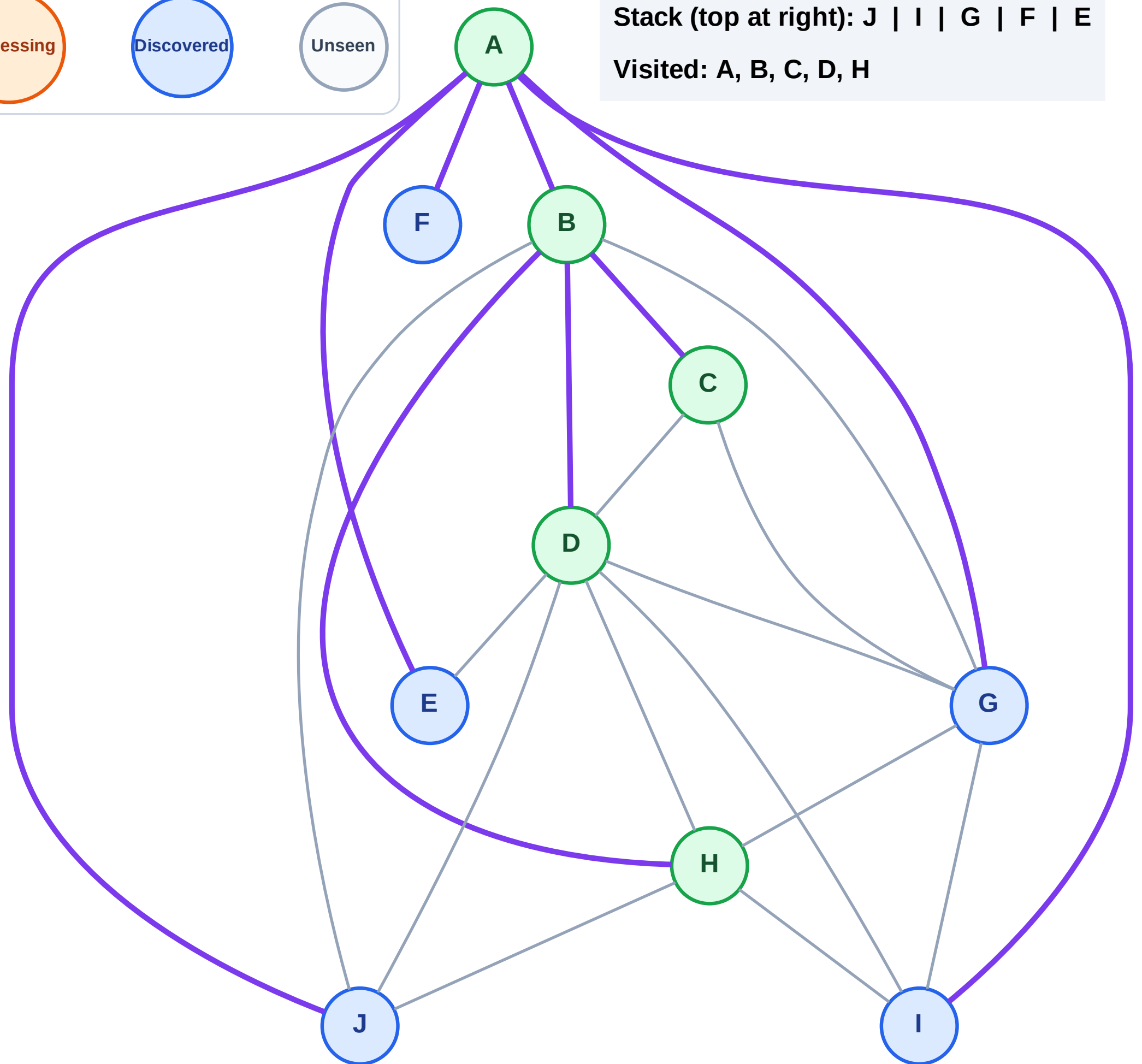
Step 11: No new neighbors from H

Legend



Stack (top at right): J | I | G | F | E

Visited: A, B, C, D, H



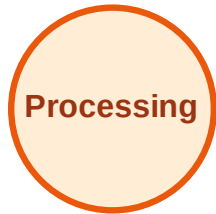
DFS Traversal

Step 12: Process node E

Legend



Complete



Processing



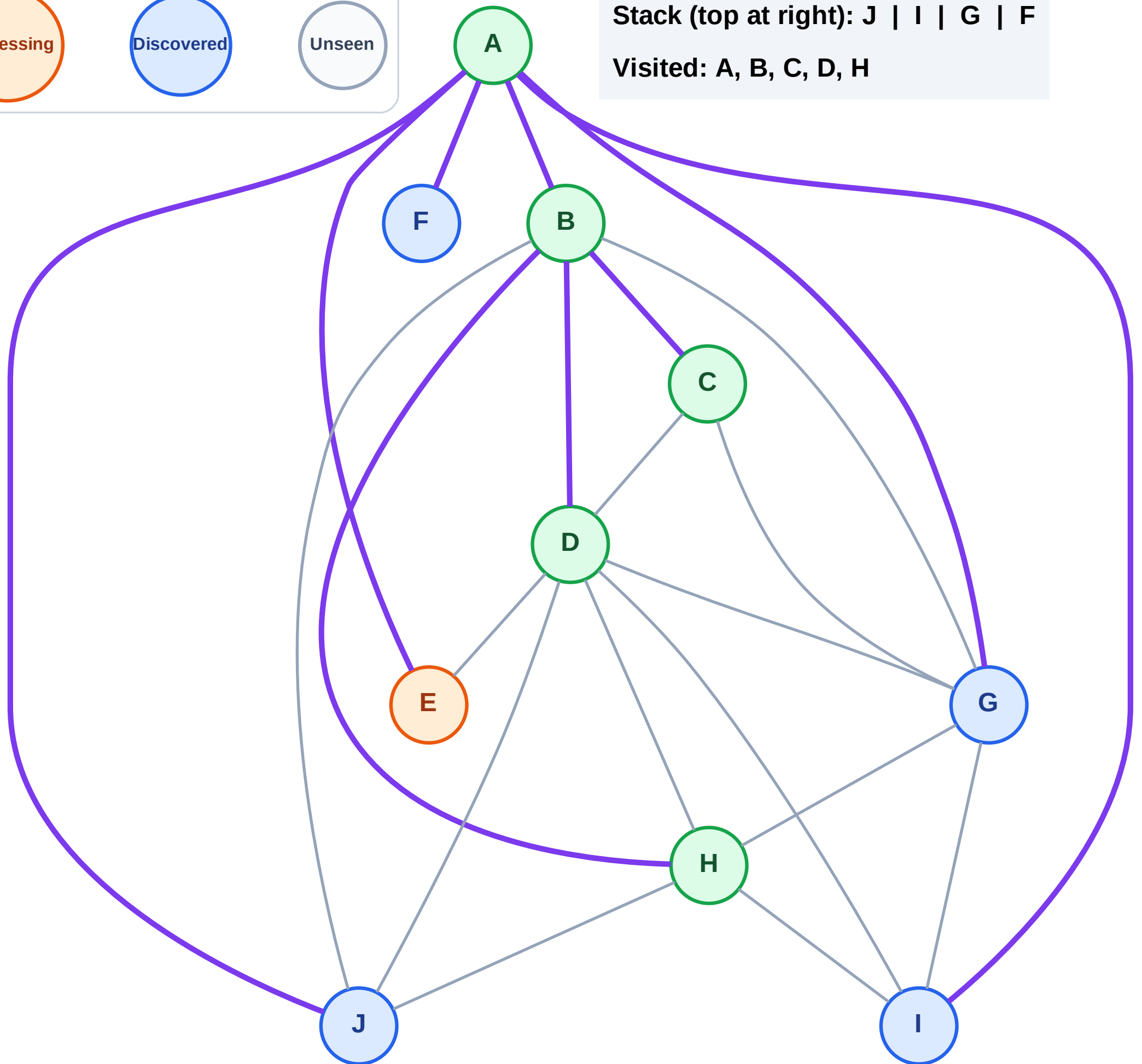
Discovered



Unseen

Stack (top at right): J | I | G | F

Visited: A, B, C, D, H



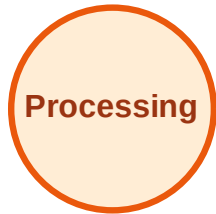
DFS Traversal

Step 13: No new neighbors from E

Legend



Complete



Processing



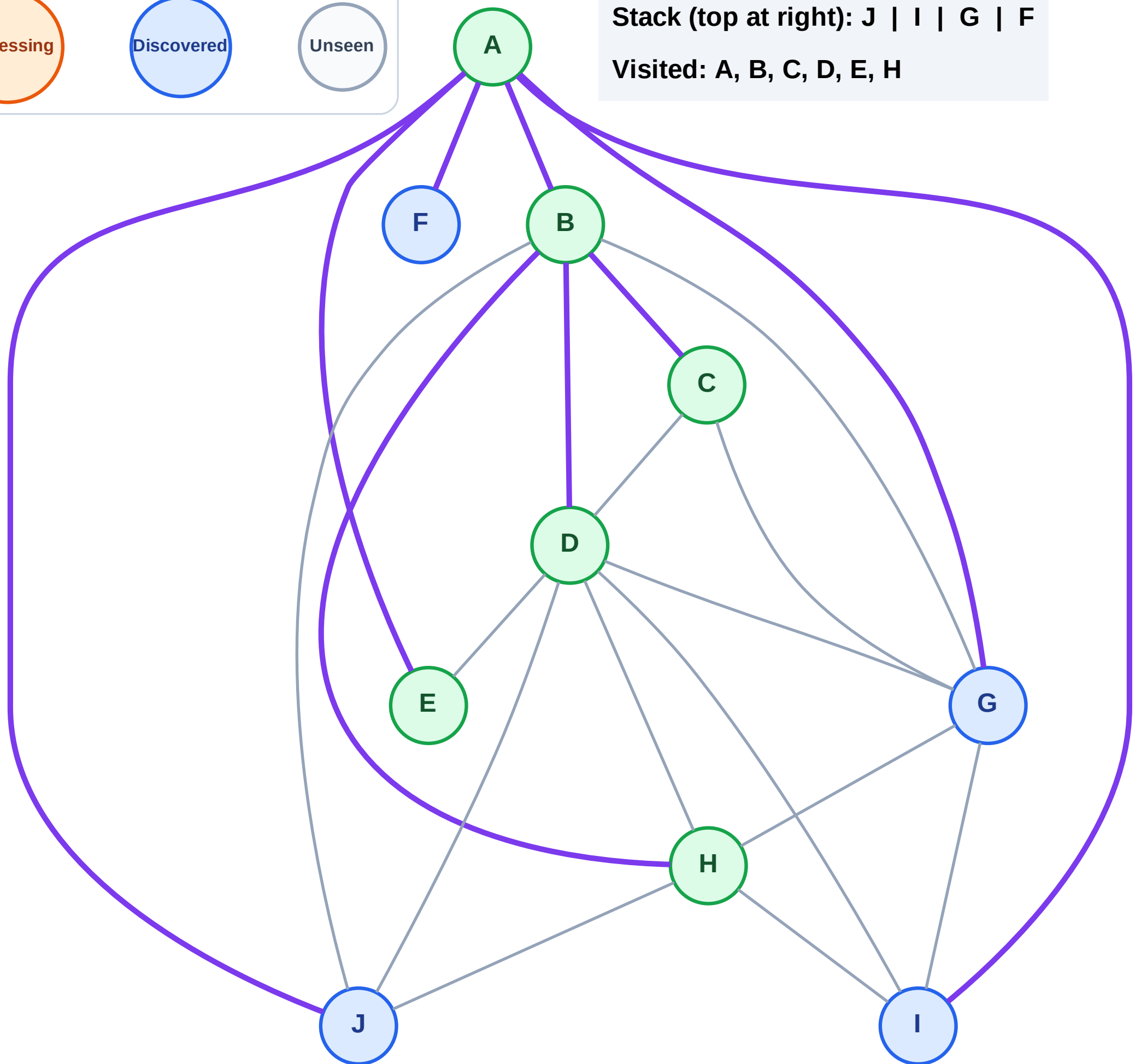
Discovered



Unseen

Stack (top at right): J | I | G | F

Visited: A, B, C, D, E, H



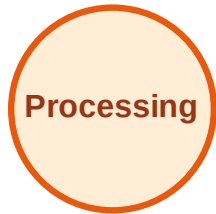
DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



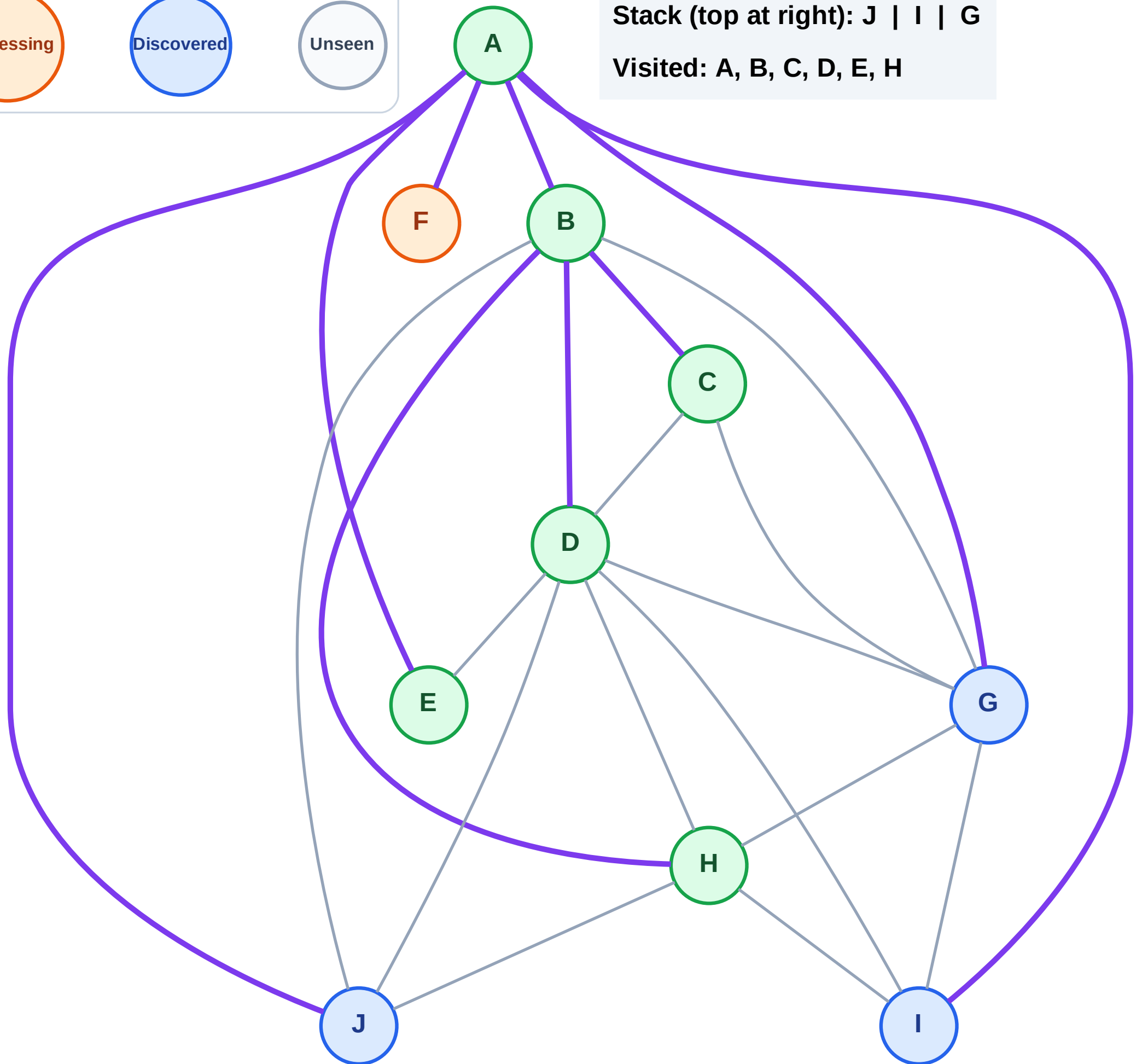
Discovered



Unseen

Stack (top at right): J | I | G

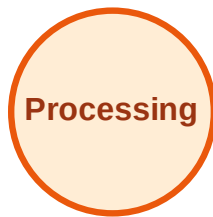
Visited: A, B, C, D, E, H



DFS Traversal

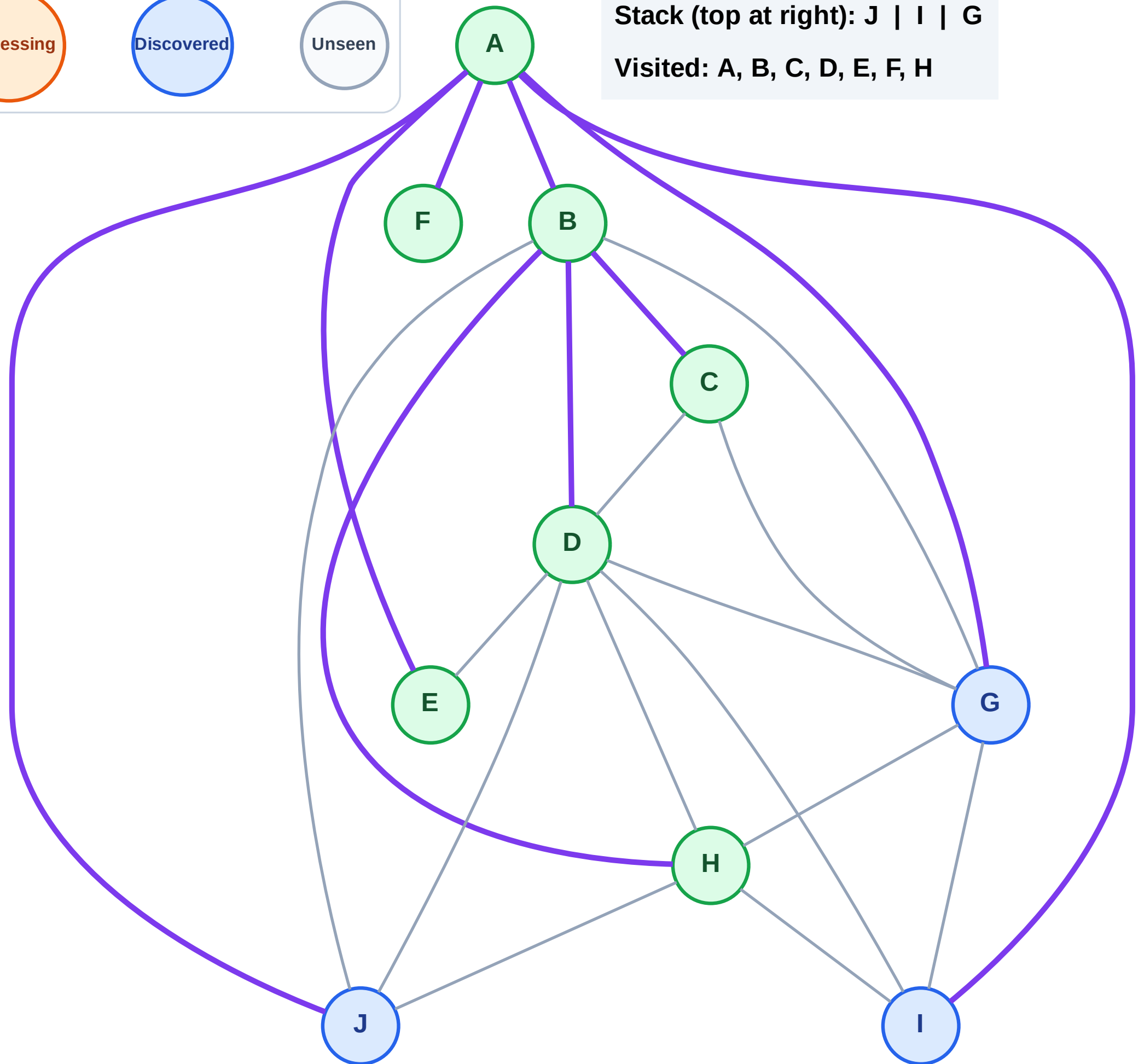
Step 15: No new neighbors from F

Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



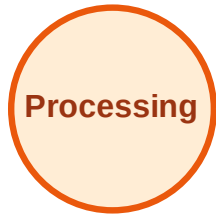
DFS Traversal

Step 16: Process node G

Legend



Complete



Processing



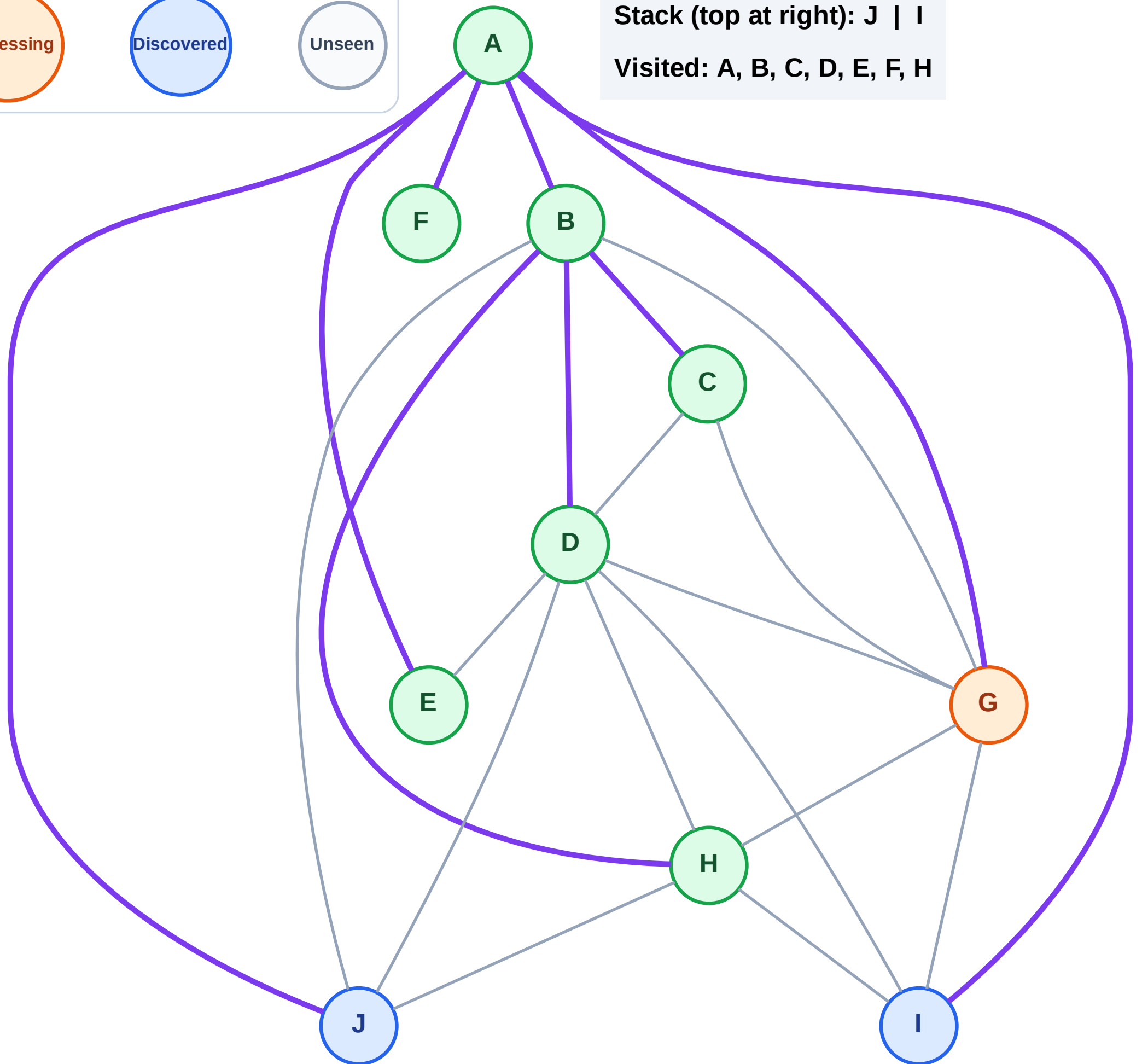
Discovered



Unseen

Stack (top at right): J | I

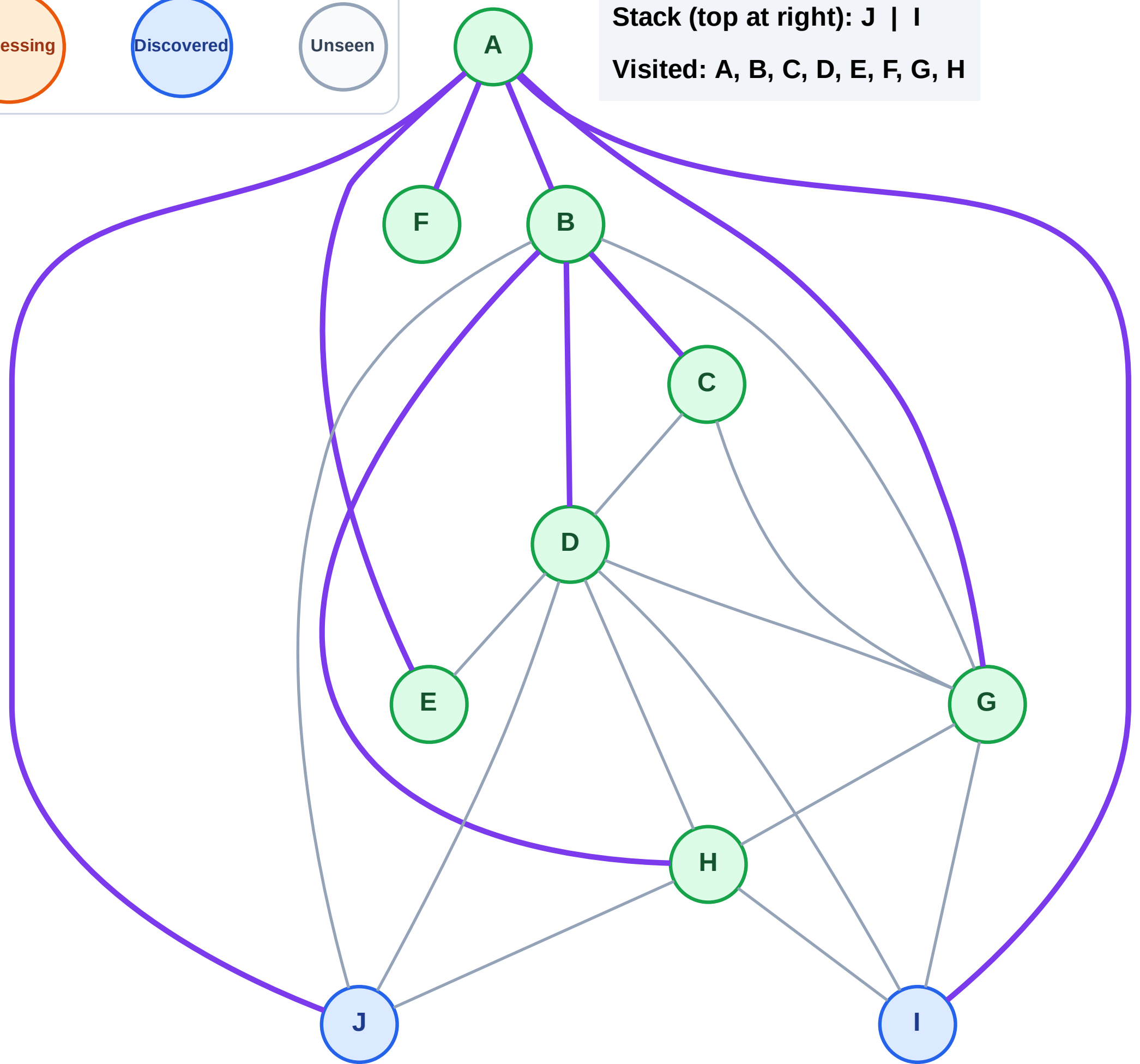
Visited: A, B, C, D, E, F, H



Step 17: No new neighbors from G

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



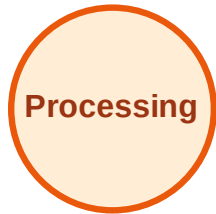
DFS Traversal

Step 18: Process node I

Legend



Complete



Processing



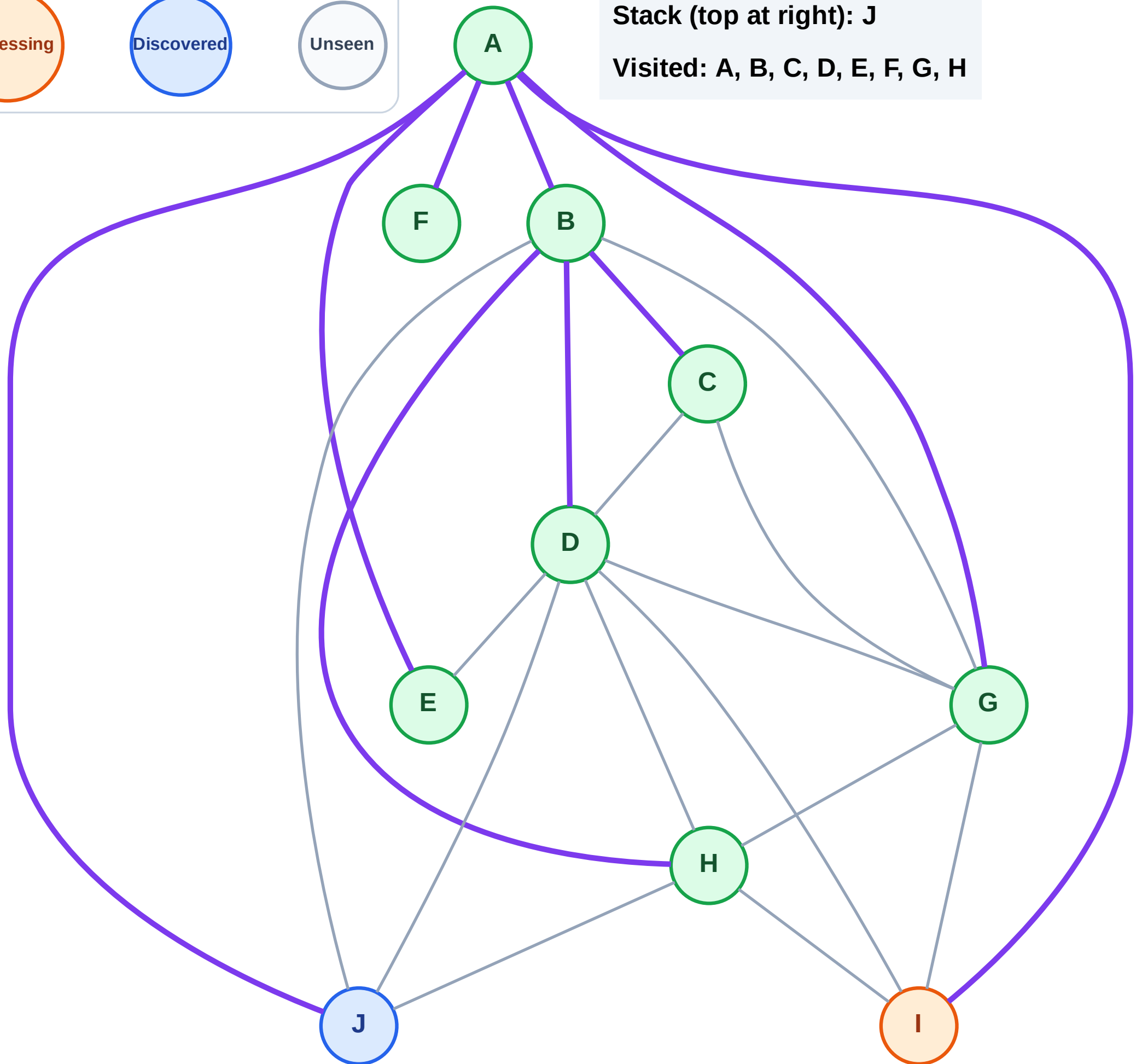
Discovered



Unseen

Stack (top at right): J

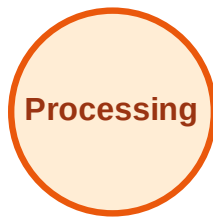
Visited: A, B, C, D, E, F, G, H



DFS Traversal

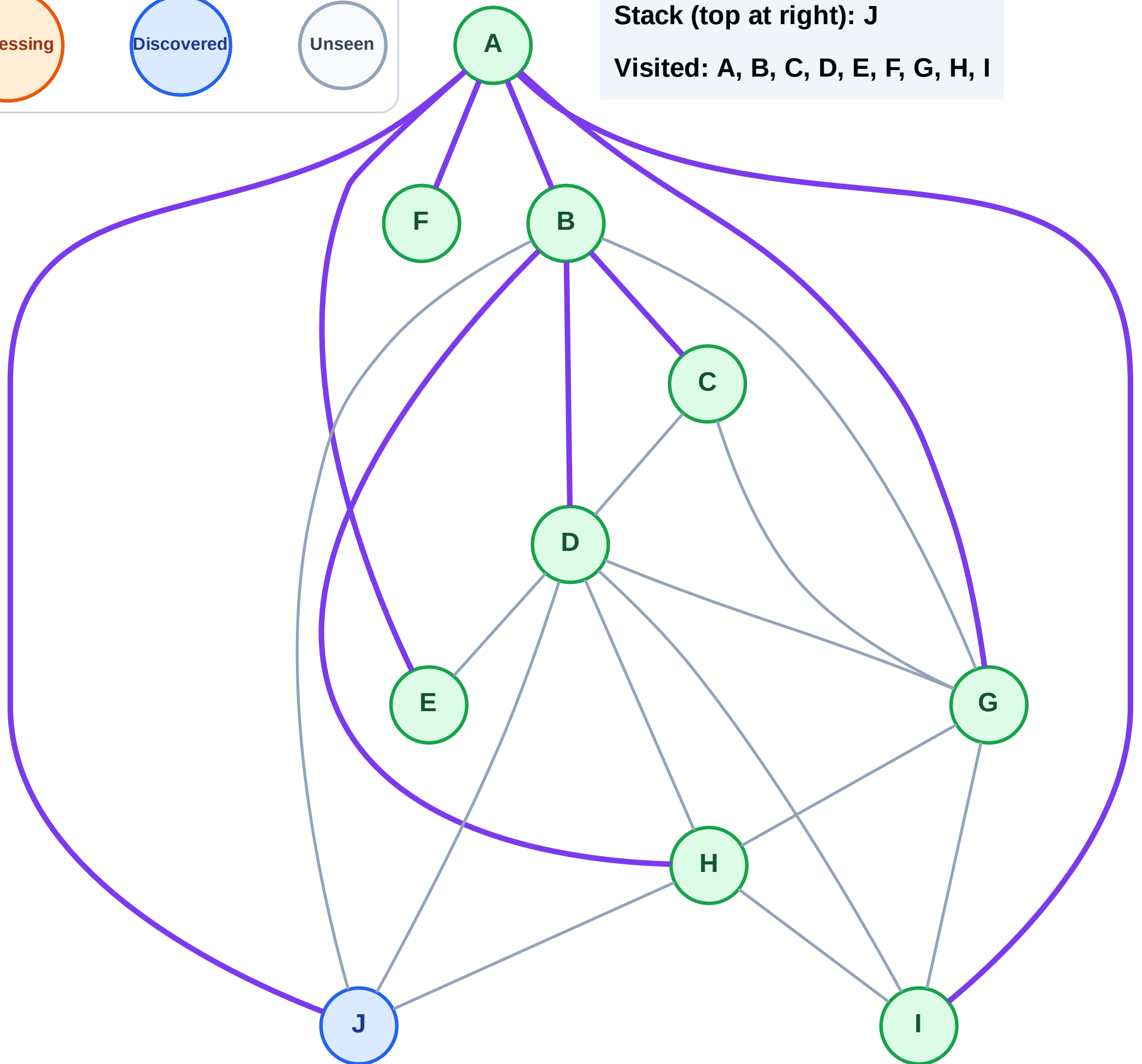
Step 19: No new neighbors from I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



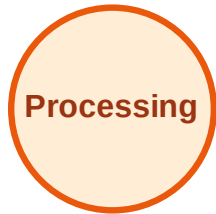
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



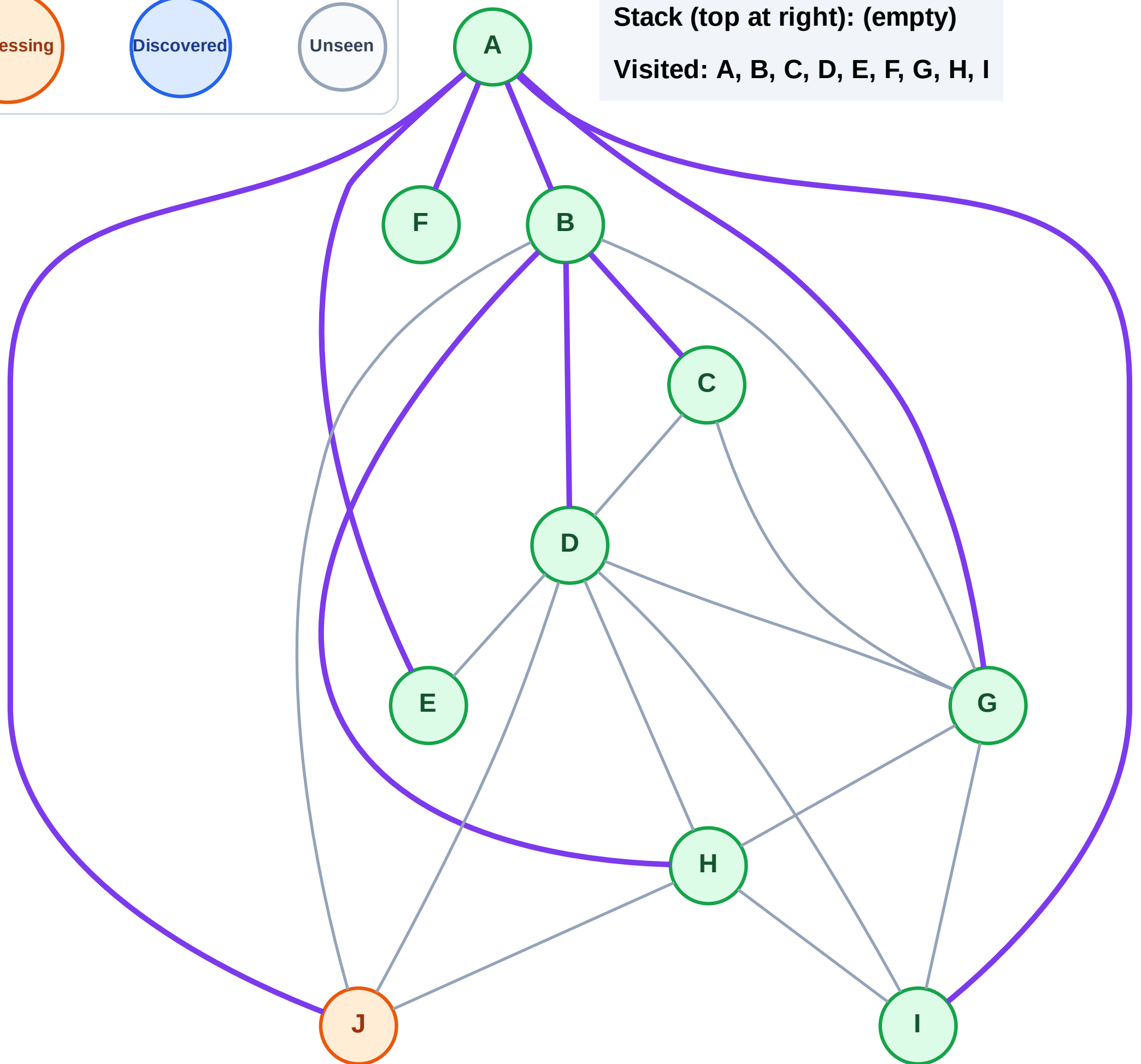
Discovered



Unseen

Stack (top at right): (empty)

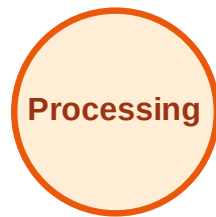
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

